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Design and Promotion of Dietary Recipes with Homologous Food and Medicine for Preventing Myopia in Jiangsu Students' Cafeteria



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Abstract:

Myopia is a major global public health problem. As early as the Eastern Han Dynasty, Chinese people discovered that *Lycium barbarum* (goji berry) could alleviate vision problems—it is a typical food–medicine-homology ingredient. We are committed to promoting the adoption of food–medicine-homology recipes in school cafeterias, introducing more such ingredients, enabling students to prevent myopia through their diet and inherit the culture of dietary therapy. We reviewed 53 books on dietary therapy and compiled 83 food–medicine-homology recipes for myopia prevention. After three questionnaire surveys with a total of 1,170 responses, the data showed that "beverages" were the most popular type of food for preventing myopia, accounting for 31.21%. We held a tasting event at Tianyi Middle School, where we personally prepared "Honey Black Sesame Milk Hot Drink" and "Goji Berry, Chenpi, Red Date Honey Drink." Compared with cola and milk tea, the homemade drinks were more nutritious and contained no additives. More than 100 cups of homemade drinks were quickly snapped up by students. Meanwhile, we also held promotional lectures at Wuxi Aier Experimental Kindergarten, Dongting Experimental Primary School, Erquan Middle School, and other schools. Students actively identified ingredients, copied recipes by hand, and asked questions—the response was enthusiastic. Recently, we reported our innovative actions to the Wuxi Sunan School Food Material Distribution Company, which serves nearly 700,000 students. Our work was highly recognized and supported by the company, and they offered suggestions for improving the food–medicine-homology recipes. The "Honey Black Sesame Milk Hot Drink" recipe was published on the company's WeChat official account, reaching a wider range of students. In the future, we will rely on the company's platform to combine food–medicine-homology recipes with outdoor exercise and eye hygiene to better prevent myopia in students.

Keywords:

Jiangsu students; Myopia prevention; Food–medicine-homology recipes; Cafeteria; Innovation action

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1 Introduction

In recent years, the global myopia rate among children and adolescents has continued to climb. The myopia rate among Chinese adolescents is already the highest in the world, and Jiangsu students' myopia rate ranks among the highest of all provinces. Myopia—a complex eye disease caused by the combined effects of genetic and environmental factors—is becoming a major public health challenge. Although various prevention strategies currently exist, each has its own limitations.

Chinese myopia dietary therapy has a long history. Our ancestors accumulated rich experience in the long struggle against eye diseases. For example, the seventh-century medical classic Essential Prescriptions Worth a Thousand Gold (Beiji Qianjin Yao Fang), compiled by Sun Simiao, records that sheep liver could treat eye ailments, and that remedy is still used in myopia dietary therapy today. At present, schools generally promote eye hygiene and healthy lifestyles, but introductions to Chinese myopia dietary therapy are relatively scarce. Meanwhile, modern students' dietary habits are increasingly fast-food-oriented, processed, and high in sugar and oil, which may lead to nutritional imbalance and even adversely affect vision. Against this background, designing scientifically sound food–medicine-homology recipes has become an important measure for schools to prevent myopia through diet. As an important place for daily meals, the school cafeteria not only meets students' basic dietary needs but also serves as a "living classroom" for nutrition education and health guidance.

As a province strong in traditional Chinese medicine, Jiangsu has unique advantages and potential in using its rich food–medicine-homology resources to prevent and improve myopia. Promoting food–medicine-homology recipes through school cafeterias can not only provide students with nutritious, vision-friendly food but also subtly spread traditional Chinese dietary therapy culture, enhancing students' health awareness and dietary literacy. The food–medicine-homology recipe program embodies the essence of traditional Chinese culture while meeting the practical needs of modern myopia prevention among students, with distinct Chinese characteristics. However, many schools in Jiangsu currently do not fully recognize the value of the food–medicine homology concept, nor are they aware of the importance of cafeterias and recipes in myopia prevention. This has led to obvious deficiencies in providing myopia-preventing nutritious meals in school cafeterias, specifically manifested as insufficient supply of green vegetables and fruits, lack of vision-improving ingredients, cooking methods that easily cause nutrient loss, and neglect of responsibility for student nutrition education.

To effectively prevent myopia, we call on relevant departments to strengthen guidance and support for school cafeterias, promote the design of myopia-preventing recipes, and incorporate them into important content of nutrition education. By designing food–medicine-homology recipes for myopia prevention, we hope to provide healthier and more scientific dietary choices for children in Jiangsu, across the country, and even worldwide, while reducing myopia risk. Through the platform of school cafeterias, we can subtly guide students to understand more vision-improving ingredients and food–medicine-homology dishes, and pay attention to the nutritional value and health benefits of food. By promoting the food–medicine homology concept and recipes, we can spread Chinese dietary therapy culture and enhance the influence and recognition of traditional Chinese dietary therapy at home and abroad.

2 Preliminary Research

2.1 Literature Review

2.1.1 Myopia prevention and control urgently needs new approaches

Holden et al. (2016) projected that by 2050, half of the world's population would be myopic, and about 10% would have high myopia. This trend poses a major public health problem; one recent review estimates that by 2055, uncorrectable vision loss in high-risk groups could rise seven- to thirteen-fold. The rise in myopia rates among children and adolescents in East Asia is particularly significant. Yin Chunjie (2022) pointed out that current main measures for controlling myopia—such as increasing outdoor activities, instilling atropine eye drops, and wearing orthokeratology lenses—all have limitations, so there is an urgent need to explore more effective prevention and control methods.

2.1.2 Myopia is correlated with diet

Gardiner (1960) proposed that diet can improve myopia to some extent. Acquired myopia mostly occurs between ages 7 and 11, and often continues to worsen until the body stops growing. A high animal protein diet is beneficial, and children who consume more vegetable protein have lower myopia progression. Ren (2022) found that the prevalence of myopia among Chinese children is related to the consumption of sugary processed foods. Lim (2010) pointed out that there is a correlation between dietary factors, myopia, and children's axial length. Masenya (2022) argued that food is the second largest factor in myopia development after outdoor exercise.

2.1.3 Government documents on myopia prevention and control

The Ministry of Education's *Implementation Plan for Comprehensive Prevention and Control of Myopia in Children and Adolescents* (2018) emphasizes the importance of nutritious meals (fish, fruits, green vegetables) for vision health and sets myopia rate control targets through 2030: children aged 6 ≤ 3%, primary school students ≤ 38%, junior high school students ≤ 60%, senior high school students ≤ 70%. The National Health Commission's *Ten Core Knowledge Points for Myopia Prevention and Control* (2023) states that ages 0–17 are a critical period for visual development and recommends consuming more fish, eggs, dairy, soy products, and fresh fruits and vegetables.

2.1.4 Feasibility of traditional Chinese dietary therapy for myopia prevention

Yao Fangwei, heir of the Yao family's traditional Chinese medicine ophthalmology, in his book *Dietary Therapy for Eye Diseases*, analyzes symptoms and causes of myopia and other conditions, and proposes 70 dietary therapy prescriptions. The Shanghai Municipal Education Commission responded positively, relying on the medical-education integration mechanism to carry out pilot work on traditional Chinese medicine appropriate techniques for myopia prevention and control, with traditional Chinese dietary therapy specifically emphasized. The Guangxi Wuzhou Municipal Education Bureau, through collecting nutritious recipes suitable for primary and secondary school students, demonstrated the feasibility of optimizing student recipes.

2.2 Questionnaire Survey

Before designing the food–medicine-homology recipes, we conducted the first survey. Through online and offline methods, we distributed questionnaires to 806 students in Jiangsu (kindergarten 27.14%, primary school 43.25%, junior high school 21.07%). The survey clarified students' current status, dietary habits, and

preferences regarding dietary myopia prevention, and evaluated the effectiveness of existing prevention strategies.

After the promotional lectures on food–medicine-homology recipes, we conducted the second questionnaire survey. For kindergarten students (1.59%), we used PPT slide reading combined with picture display and hand-raising responses. For primary school (8.6%), junior high school (51.91%), and senior high school (37.58%) students, we used on-site answering. The second survey assessed the effectiveness of earlier publicity by asking respondents about their awareness of the "food–medicine homology" concept. The questionnaire also investigated respondents' acceptance and willingness to purchase myopia-preventing food–medicine-homology recipes, their evaluation of the scientific soundness of the recipes, and acceptable price positioning. At the tasting event for handmade myopia-preventing drinks, we conducted the third survey. Through small questionnaires attached to flyers, we learned about the taste and acceptance of the homemade myopia-preventing food–medicine-homology drinks.

As can be seen from the comparison of myopia status and distress levels between the two questionnaire surveys (Figure 1), the vision status of the surveyed group deteriorated significantly during the period between the two surveys: the proportion of those without myopia plummeted from 43.62% to 21.97%, and the myopic population increased substantially. Meanwhile, the “distress” among the myopic population intensified markedly: the proportion of those who were myopic and felt distressed soared from 10.57% to 32.96%, while those who did not feel distressed rose only slightly from 37.26% to 38.22%, indicating that more than 30% of respondents were not only myopic but also anxious about it.

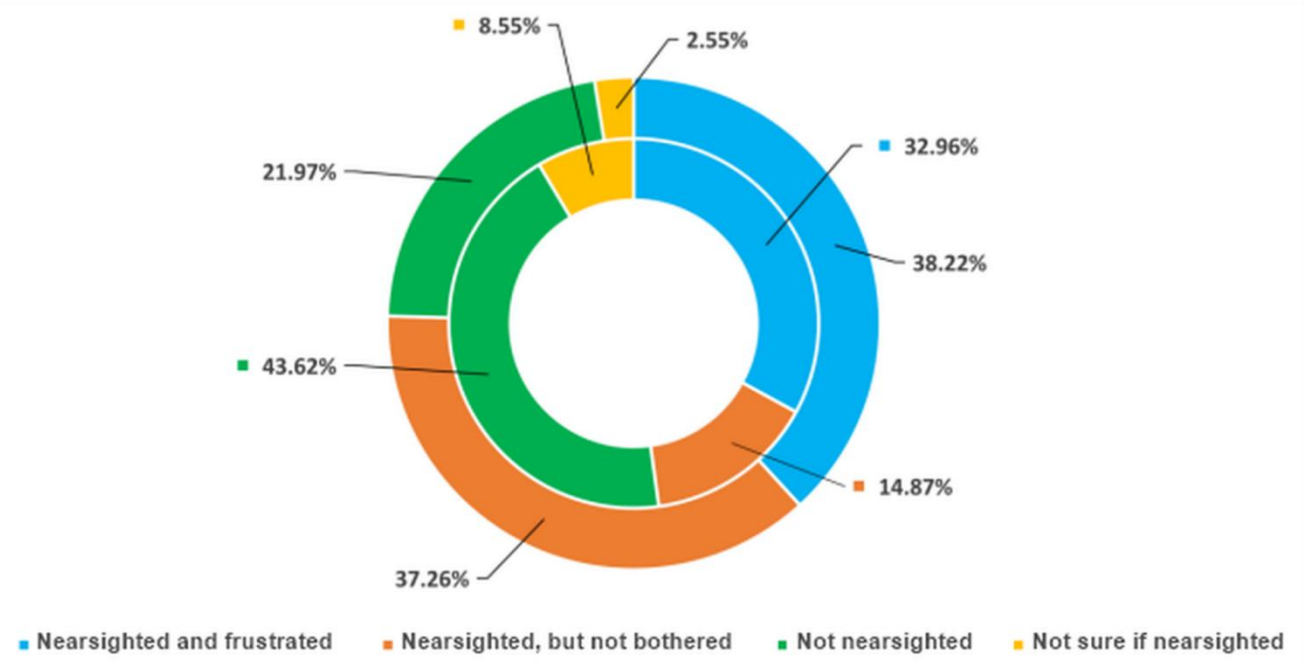


Figure 1 The comparison of myopia status and distress levels between the two questionnaire surveys

Regarding the impact of myopia on students, the total myopia rate among respondents was 55.57%. Across the two surveys, the average proportion of students who were myopic and troubled was 34.43%, which deserves sufficient attention. The proportion of students without myopia dropped significantly from 43.62% in the first survey to 21.97% in the second survey. This indicates that as school years advance, students face increased academic pressure and prolonged eye use, which may lead to increased myopia incidence.

3 Innovation Actions

3.1 Design of Myopia-Preventing Food–Medicine-Homology Recipes and Compilation of Prevention Mechanisms

We consulted 53 formally published books and collected and organized "108 recipes." After review by dietary therapy experts and culinary nutrition professionals, 83 dishes were selected that can be directly used in cafeterias. We also excerpted the mechanisms by which food–medicine-homology ingredients prevent myopia from the books, sorted out various ingredients and their nutrient types, and compiled a popular science Q&A called "55 Questions on Dietary Therapy Knowledge for Myopia Prevention." The recipes and "55 Questions" were printed as 2 roll-up banners and five posters, placed in schools or school gates where students and parents gather.

3.2 Publicity and Promotion of Food–Medicine-Homology Recipes

We selected dishes with cultural allusions for lectures—for example, "Goji Leaf and Egg Soup" includes poetry by Su Dongpo praising goji berries. We adopted different formats for students of different age groups and held promotional lectures at multiple schools in Wuxi, such as Aier Experimental Kindergarten, Dongting Experimental Primary School, and Erquan Middle School. We produced myopia-preventing dietary therapy knowledge posters and bulletin boards, erected in school cafeterias for students to study. We also cooperated with Tianyi Middle School cafeteria to make two homemade myopia-preventing food–medicine-homology drinks, organized tasting events, and collected feedback.

3.3 Empirical Application of Food–Medicine-Homology Recipes

We collected recipes from multiple kindergartens online and, after some effort, obtained another kindergarten's recipes and vision data. We used Excel tables to compare and analyze the recipes of two kindergartens, looking for differences in nutrient intake and directions for optimization. We also contacted China's largest food material distribution company—Wuxi Sunan School Food Material Distribution Company. This is a company serving nearly 700,000 students in Wuxi. Our team reported the survey and promotion results to the company's nutrition experts and received suggestions for improving food–medicine-homology recipes. Sunan Company is highly recognized and actively supports our research and will provide a long-term practical platform for our future promotion activities.

4 Results Analysis

4.1 Innovation Analysis

4.1.1 Established design principles for food–medicine-homology recipes

In the first survey, data on students' awareness of cafeteria recipes showed that 78.69% of students believed lutein intake was the lowest, and vitamin A, C, and anthocyanins were also considered insufficient by many students (each over 57%). When designing recipes, attention should be paid to increasing vegetables and fruits rich in vitamin C, anthocyanins, carotene, zinc, and other nutrients, and introducing vision-improving ingredients from the food–medicine homology list, such as goji berries and chrysanthemum, reflecting the food–medicine homology concept.

4.1.2 Empirical analysis linking recipes with vision data

We searched for school recipes online and used the obtained recipe and vision data from another kindergarten to conduct empirical research on nutrient intake using cafeteria recipes from two kindergartens (referred to as "Kindergarten 1" and "Kindergarten 2"). The comparative analysis results are as follows:

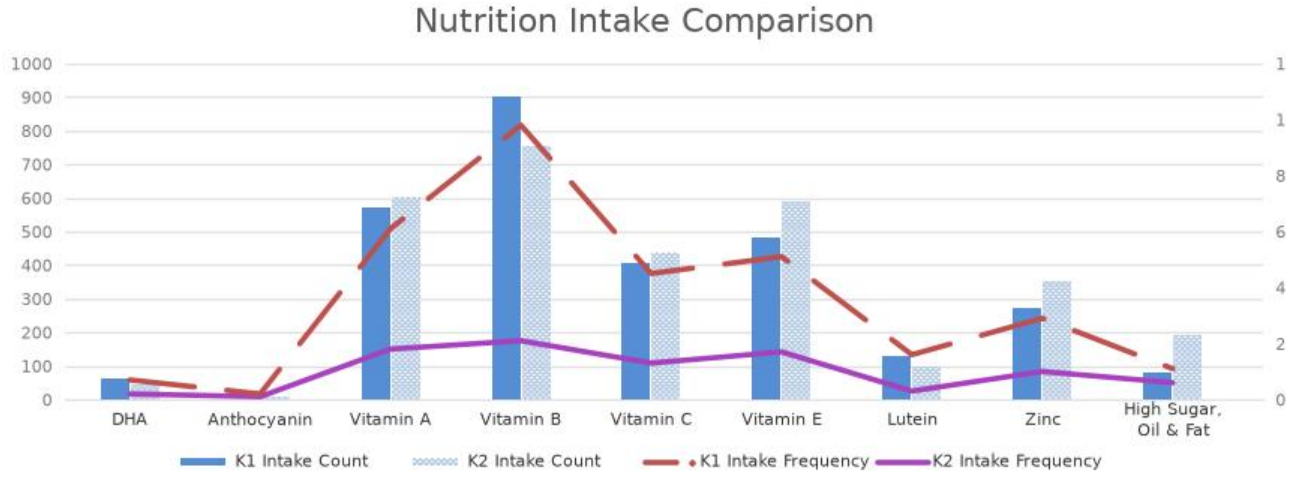


Figure 2 Comparison of Nutritional Intake Based on Menu Analysis of Kindergarten 1 and Kindergarten 2

We conducted frequency ratio analysis on the daily recipes of Kindergarten 1 and Kindergarten 2, extracted 50 ingredients, classified them into 9 nutrient categories, including DHA and anthocyanins, and produced a nutrient intake comparison chart. Kindergarten 1 had higher intake ratios for most vision-related ingredients. For vitamin B intake frequency, Kindergarten 1 was 9.826087, and Kindergarten 2 was 2.173913—a significant decrease.

For other nutrient frequency ratios, Kindergarten 2 was also significantly lower than Kindergarten 1. For high-sugar, high-oil, and high-fat food intake, Kindergarten 1 had a frequency of 0.902174, and Kindergarten 2 had 0.572464—Kindergarten 1 was relatively higher, and excessive intake may affect young children's health and vision. Kindergarten 2 should increase the supply of vision-beneficial ingredients and redesign its recipes to ensure nutritional balance. Facts prove that there are certain differences in nutrient intake in school recipes design.

4.2 Feasibility Analysis

4.2.1 Safety Guarantee of Recipe Design

The first survey showed that regarding expert recommendation needs, 64.31% of students preferred recommendations from Western medicine ophthalmologists; traditional Chinese medicine ophthalmologists (16.73%) and nutritionists (13.51%) had lower support; chefs (0.87%) and education experts (1.86%) had low recognition. Our team believes that guidance from traditional Chinese medicine ophthalmologists, nutrition experts, and cafeteria chefs should be strengthened. We had the "108 recipes" reviewed by a dietary therapy doctor from the Traditional Chinese Medicine Department of Zhenjiang Integrated Traditional Chinese and Western Medicine Hospital and a culinary nutrition professor from the School of Tourism and Culinary Arts at Wuxi Vocational Institute of Commerce. Safety was confirmed, increasing trust among teachers and students.

4.2.2 Student Recognition and Support from School Cafeteria Departments

(1) Students have a positive attitude toward myopia-preventing food–medicine-homology foods. The first survey showed that 44.24% were very willing to purchase, and 33.33% were somewhat willing. After the

second survey refined food types, the proportion unwilling to purchase was only 1.59%. Both surveys indicated that most students held a positive attitude and were willing or somewhat willing to purchase.

(2) Students are willing to pay reasonable prices. In the first survey, the 10–20 yuan per meal price range was most popular (46.72%). In the second survey, prices equal to cafeteria prices (50.64%) or slightly higher (24.2%) were widely accepted. In the first survey, 27.38% of students accepted dishes over 20 yuan; in the second survey, 20.06% of students indicated that price was not the main consideration—the key was effectiveness.

(3) Tasting events were successful. Both surveys highlighted the importance of tasting events (first survey 68.9%; second survey 60.83%). Our team organized a tasting at Tianyi Middle School. Two homemade food–medicine-homology drinks (Tables 1 and 2) were delicious. The Goji Berry, Chenpi, Red Date Honey Drink—40 cups were gone in 2 minutes; the Honey Black Sesame Milk Hot Drink—30 minutes to sell out. An on-site survey showed that over 95% of students filled out the third questionnaire or responded in conversation, with Honey Black Sesame Milk being more popular. Nutritional analysis showed that both drinks had no additives, and their vision-related nutrients and mineral content were superior to milk tea and cola (Tables 3 and 4). Honey is a natural food–medicine-homology ingredient, making the drinks healthier.

Table 1 Nutritional Composition Analysis of Honey Black Sesame Milk Hot Drink

Preparation Ingredients:		
Ingredient	Weight	Unit
Pure milk	1000	ml
Roasted black sesame	50	g
Honey	50	g
Preparation Method:		
1. First, add 1000 ml of milk to a blender or soy milk maker; 2. Then add 50 g ground roasted black sesame and 50 g honey; 3. Use the soy milk mode to blend and heat; once cooked, it is ready to drink.		
Note: Food–medicine-homology ingredients include black sesame and honey.		
Nutrient Analysis:		
Nutrient	Intake	Unit
Protein (g)	44.75	g
Energy (kJ)	3809	kJ
Calcium (mg)	1502	mg
Iron (mg)	13.85	mg
Zinc (mg)	5.25	mg
Phosphorus (mg)	1319.5	mg
Vitamin A (μg)	450	μg
Thiamine B1 (mg)	0.53	mg
Riboflavin B2 (mg)	1.75	mg
Niacin (mg)	3.7	mg
Ascorbic acid/C (mg)	1.5	mg
Main Nutrient Ratio Relationships:		
Nutrient	Requirement %	Actual %
Protein	8–20	19.43%
Fat	20–30	38.10%
Animal fat	30–50	38.46%
Carbohydrate	50–65	42.47%

Table 2 Nutritional Composition Analysis of Goji Berry, Chenpi, Red Date Honey Drink

Preparation Ingredients:		
Ingredient	Weight	Unit
Goji berries	40	g
Jujubes	100	g
Chenpi (dried tangerine peel)	20	g
Honey	50	g
Purified water	2000	ml
Preparation Method:		
1. First, add 1000 ml of purified water to a pot, then add 40 g of goji berries, 100 g of jujubes (washed and cut open), 20 g of chenpi (washed), and 50 g of honey in sequence; 2. After bringing to a boil over high heat, simmer over low heat for 20 minutes; strain the decoction and set aside; 3. Add another 1000 ml of purified water to the pot, bring to a boil, then simmer for 20 minutes; strain and pour into the first batch of decoction, stir well, and it is ready to drink.		
Note: Food–medicine-homology ingredients include goji berries, jujubes, chenpi, and honey.		
Nutrient Analysis:		
Nutrient	Intake	Unit
Protein (g)	9.46	g
Energy (kJ)	2663	kJ
Calcium (mg)	96.4	mg
Iron (mg)	6.4	mg
Zinc (mg)	1.427	mg
Phosphorus (mg)	136.1	mg
Vitamin A (μg)	720	μg
Thiamine B1 (mg)	0.22	mg
Riboflavin B2 (mg)	0.359	mg
Niacin (mg)	3.65	mg

Table 3 Nutritional Composition Analysis of a Well-Known Brand of Taro-Flavored Milk Tea

Ingredients:		
Milk tea powder ingredients: non-dairy creamer (glucose syrup, refined vegetable oil, sodium caseinate, mono- and diglycerides of fatty acids, dipotassium hydrogen phosphate, silicon dioxide), white granulated sugar, edible glucose, skimmed milk powder, instant black tea powder, instant oolong tea powder, whey protein powder, food additives (tricalcium phosphate), edible flavors and fragrances. Milk tea fruit ingredients: water, coconut fiber, high-fructose corn syrup, white granulated sugar, edible glucose, food additives (citric acid, sodium citrate, sodium carboxymethyl cellulose, guar gum, calcium lactate, potassium sorbate).		
Nutrient Analysis:		
Nutrient	Intake (per 80 g total)	Unit
Energy	901	kJ
Protein	1.2	g
Fat	7.5	g
Carbohydrate	35.5	g
Sodium	67	mg

Table 4 Nutritional Composition Analysis of a Well-Known Brand of Cola

Ingredients:		
Main ingredients: water, high-fructose corn syrup, white granulated sugar, food additives (carbon dioxide, caramel color, phosphoric acid, caffeine), edible flavors, and fragrances.		
Nutrient Analysis:		
Nutrient	Intake (per 100 ml)	Unit
Energy	180	kJ
Protein	0	g
Fat	0	g
Carbohydrate	10.6	g
Sugars	10.6	g
Sodium	12	mg

4.3 Effectiveness Analysis

4.3.1 Lectures received enthusiastic responses

Both surveys showed that students have a strong demand for nutritional knowledge about myopia-preventing dishes. In the first survey, 69.64% of students hoped recommendations would include detailed nutritional knowledge. Lectures were welcomed by students: first survey 60.72%, second survey 19.11%, indicating that providing relevant knowledge is crucial for promoting students' acceptance of new recipes. Our team held food–medicine-homology recipe promotion lectures at multiple schools, which were warmly welcomed by students. Students showed great interest—some actively participated in interactions, and some copied recipes by hand to take home.

Two questionnaire surveys found that students' awareness of the "food–medicine homology" concept significantly improved, and most were willing to accept food–medicine-homology dishes and hoped to see them in school cafeterias. Regarding the effectiveness of the presentations, the second survey showed that most students could understand the myopia-preventing effects of natural ingredients through the presentations, with "very knowledgeable" at 28.34%, "somewhat knowledgeable" at 30.25%, and "know a little" at 30.89%. Compared with the first survey, awareness of the "food–medicine homology" concept increased from 54.28% to 57.96%, thanks to targeted publicity through lectures. Continuous knowledge popularization and effective publicity are expected to improve students' eye health awareness and myopia prevention outcomes.

4.3.2 Promotional posters had significant effects

The first survey's Question 13 showed that traditional materials such as posters and brochures were influential in both surveys (50.31%), suggesting that promotion needs to comprehensively consider multiple methods. Based on this, our team printed the "108 recipes" and produced 2 roll-up banners placed at school gates and cafeterias; we also compiled "55 Questions on Dietary Therapy" and printed them as five bulletin boards, erected in areas where students and parents gather. The posters attracted teachers and students—during breaks, students often stopped to inquire about ingredient names and effects, and parents also offered optimization suggestions when picking up their children.

4.3.3 Sunan Company's introduction of food–medicine-homology recipes

Wuxi Sunan Food Material Company highly recognized and supported our research, believing that student teams can be relied upon to promote healthy food, and promised to provide a long-term practical platform. We analyzed the existing recipes that Sunan provides to school cafeterias and added food–medicine-homology ingredients with myopia-preventing effects. At the same time, we conducted nutritional analysis on the new food–medicine-homology recipes to evaluate their actual effects.

The first survey showed that only 37.79% of students believed the cafeteria often provided myopia-preventing food; some students thought it was sometimes, rarely, or never provided, and 22.68% of students were unsure. Meanwhile, most students were satisfied or somewhat satisfied with the nutritional balance of cafeteria dishes (29.24% and 38.54%, respectively), but there was still room for improvement (20.94% thought it was average, 8.43% and 2.85% were dissatisfied). Therefore, the original recipe "Recipe 1" shown in Table 5 of Sunan Food Material Company met nutritional standards but was relatively weak in myopia-prevention targeting and innovation, and some dishes had high oil and seasoning content and lacked hard foods (see shaded dishes in Table 5).

In addition, we cooperated with Wuxi Sunan School Food Material Distribution Company to integrate food–medicine-homology recipes into their food material distribution services and improved the existing recipes. The improved recipe from Sunan Company, shown as "Recipe 2" in Table 6, introduces myopia-preventing ingredients and plans to pilot in one or two schools in March 2025. The features of "Recipe 2" are as follows:

(1) Focus on increasing fruits and vegetables. The survey showed that students generally believed that ingredients rich in lutein and other nutrients were insufficiently consumed (lutein 78.69%, etc.), and hoped myopia-preventing dishes would include more fruits (87.98%), vegetables (78.81%), and meat/egg items (49.69%). The new recipe, "Recipe 2" enriches these ingredients, such as apples, ugly oranges, and West Lake water shield, increasing interest and innovation.

(2) Introducing vision-improving ingredients, such as blueberries, purple sweet potatoes, and oats, rich in vitamin E, anthocyanins, and other nutrients, with clear effects on myopia prevention.

(3) Emphasizing healthy cooking methods. Such as steamed purple sweet potatoes and millet, yam, and goji berry porridge, preserving ingredients' nutrients and reducing the adverse effects of high temperature on food.

In addition, the "Honey Black Sesame Milk" recipe written by our team was published on Sunan Company's official account. Within just three days, it received several hundred views. In the two surveys, social media usage was not high (first survey 27.88%, second survey 7.96%), mainly because students' access to mobile phones and computers is restricted. However, the influence of social media among young students, especially high school students, should not be ignored. As social media becomes more widespread, its importance may gradually increase. Therefore, our team produced three short videos recommending the food–medicine-homology recipe "Goji Leaf and Egg Soup," introducing traditional Chinese medicine ingredients, the food–medicine homology concept, modern nutritional analysis, and the dish's historical allusions and preparation methods. We then uploaded these short videos to Tianyi Middle School's official account, Douyin, Xiaohongshu, and other social media platforms for promotion.

Table 5 Improved Recipe Recipe 2 of Wuxi Sunan School Food Material Distribution Co., Ltd.

Category	Monday	Tuesday	Wednesday	Thursday	Friday
Morning Snack	Pure milk Edible portion: pure milk: 125 Black sesame flower roll Edible portion: sesame seeds: 1.5 Edible portion: wheat flour: 10	Pure milk Edible portion: pure milk: 125 Crispy biscuits Edible portion: biscuits: 12	Pure milk Edible portion: pure milk: 125 Steamed purple sweet potato Edible portion: purple sweet potato: 60	Pure milk Edible portion: pure milk: 125 Soda biscuits Edible portion: biscuits: 12	Pure milk Edible portion: pure milk: 125 Steamed corn Market product: corn: 60
Lunch	Edible portion: refined salt: 1 Edible portion: oil: 6.5 Oat rice Edible portion: oats: 10 Edible portion: japonica rice: 45 Braised chicken wings Market product: chicken wings: 70 Stir-fried water shield with fried tofu Edible portion: water shield Edible portion: fried tofu: 5 Spinach and pork liver soup Edible portion: spinach: 15 Edible portion: pork liver: 15 Edible portion: carrot: 5	Edible portion: refined salt: 1 Edible portion: oil: 6.5 Soft rice Edible portion: japonica rice: 55 Braised sea bream Market product: sea bream: 50 Stir-fried celery with dried tofu Edible portion: dried tofu: 25 Edible portion: celery: 40 Tomato and egg soup Edible portion: milk tomato: 30 Market product: egg: 10 Small pear soup Market product: pear: 90 Market product: white fungus: 0.2 Edible portion: goji berries: 1.5 Market product: jujube: 5	Edible portion: refined salt: 1 Edible portion: oil: 6.5 Carrot rice Edible portion: japonica rice: 45 Edible portion: carrot: 10 Sweet and sour pork tenderloin strips Market product: pork: 45 Stir-fried seasonal vegetables with double cauliflower Edible portion: white cauliflower: 30 Edible portion: bell pepper: 15 Edible portion: broccoli: 30 Chicken feather greens, tofu skin, and egg soup Market product: egg: 10 Edible portion: tofu skin: 1 Edible portion: chicken feather greens: 12	Edible portion: refined salt: 1 Edible portion: oil: 6.5 Soft rice Edible portion: japonica rice: 55 Potato stewed beef Edible portion: potato: 10 Market product: beef: 45 Edible portion: carrot: 20 Stir-fried crown daisy Edible portion: crown daisy: 75 Lettuce and wood ear egg soup Edible portion: dried black fungus: 0.5 Edible portion: lettuce: 25 Market product: egg: 10	Edible portion: refined salt: 1 Edible portion: oil: 6.5 Tomato meat sauce Italian pasta Edible portion: noodles: 40 Edible portion: onion: 15 Edible portion: milk tomato: 40 Market product: pork: 20 Radish and pork rib soup Market product: pork ribs: 20 Edible portion: white radish: 30
Afternoon Snack	Beef vermicelli soup Edible portion: oyster mushroom: 10 Edible portion: baby bok choy: 10 Edible portion: vermicelli: 12 Edible portion: beef: 9 Ponkan Market product: ugly orange: 100	Daily nuts Edible portion: daily nuts: 12.5 Beibei pumpkin Market product: pumpkin: 60 Small pear soup Market product: pear: 90 Market product: white fungus: 0.2 Edible portion: goji berries: 1.5 Market product: jujube: 5 Daily nuts Edible portion: daily nuts: 12.5 Beibei pumpkin Market product: pumpkin: 60	Three fresh diced noodles Market product: pork: 9 Edible portion: noodles: 25 Edible portion: milk tomato: 20 Edible portion: lettuce: 10 Cherry tomatoes Market product: cherry tomato: 80 Millet, yam, goji berry porridge Edible portion: millet: 15 Edible portion: goji berries: 1.5 Market product: yam: 25	Blueberries Edible portion: blueberries: 40 Mousse-flavored yogurt Edible portion: yogurt: 125 Handmade sponge cake Edible portion: sponge cake: 40	Apple Market product: apple: 90

Table 6 Improved Recipe

by Wuxi Sunan School Food Ingredient Distribution Co., Ltd.: Recipe 2

Category	Monday			Tuesday			Wednesday			Thursday			Friday		
	Item	Ingredient	Weight (g)	Item	Ingredient	Weight (g)	Item	Ingredient	Weight (g)	Item	Ingredient	Weight (g)	Item	Ingredient	Weight (g)
Student Snack	**Pure Milk**	Pure Milk, 125 ml	1	**Milk-flavored Dinner Roll**	Milk-flavored Dinner Roll, 40 g/piece	1	**Pure Milk**	Pure Milk, 125 ml	1	**Pumpkin Seed Kernel Cake**	Pumpkin Seed Kernel Cake, 40 g/piece	1	**Banana**	Imported Banana (piece)	1
Staple Food	**Soft Steamed Rice**	Japonica Rice	120	**Corn Grits Rice**	Corn Grits, 150 g/pack	3	**Soft Steamed Rice**	Japonica Rice	120	**Oat Rice**	Rolled Oats, 150 g/pack	3	**Soft Steamed Rice**	Japonica Rice	120
					Japonica Rice	117					Japonica Rice	117			
Main Meat Dish	**Braised Duck Leg Chunks**	Chilled Fresh Duck Leg Chunks	150	**Clear-Simmered / Braised Pork Blade Bone Chunks**	Chilled Fresh Pork Blade Bone Chunks	145	**Curry Beef Chunks**	Domestic Chilled Fresh Beef Chunks	100	**Stir-fried Shrimp with Seasonal Vegetables**	Large Bell Pepper Chunks	30	**Steamed Pork Patty with Egg**	Chilled Fresh Minced Pork	70
								New Potato Chunks	30		Carrot Chunks	20			
											Chilled Fresh Peeled Shrimp	50			
								Carrot Chunks	15		Egg (cleaned)	5		Egg (cleaned)	40
											Onion Shreds	15			
Side Meat Dish	**Sliced Pork with Cauliflower and Carrot**	Cauliflower Florets (loose-curd 'songhua' cauliflower)	60	**Scrambled Eggs with Tomato**	Tomato	90	**Sweet and Sour Pork Tenderloin Strips**	Pork	45	**Chicken Drumettes in Plum Sauce**	Chilled Fresh Chicken Drumettes (50-60 g/piece)	120	**Stir-fried Celtuce with Pork Slices**	Celtuce (stem lettuce) Slices	70
		Chilled Fresh Lean Pork Slices	40		Egg (cleaned)	45								Chilled Fresh Lean Pork Slices	40
		Carrot Slices	10												
Vegetable Dish	**Spinach with Thin Tofu Sheets**	Spinach Cuts	85	**Radish in Scallion Oil**	White Radish (Daikon) Chunks	105	**Baby Napa Cabbage with Shimeji Mushrooms**	Baby Napa Cabbage Cuts	90	**Maobai Greens with Fried Tofu Triangles**	Maobai Greens (leafy Chinese cabbage) Cuts	100	**Braised Glass Noodles**	Chilled Fresh Minced Pork	5
		Thin Tofu Sheets (bean curd sheets)	5					Shimeji (Brown Beech Mushroom)	5		Fried Tofu Triangles	5		Sweet Potato Glass Noodles, 500 g/pack	18
Soup	**Egg Sour Non-spicy Soup**	Egg (cleaned)	15	**Napa Cabbage and Shredded Pork Soup**	Napa Cabbage Cuts	25	**Minced Pork, Shiitake and Tofu Soup**	Soft Tofu	20	**West Lake Beef Thick Soup (Geng)**	Domestic Chilled Fresh Minced Beef	5	**Meaty Pork Bone and Radish Soup**	Chilled Fresh Meaty Pork Back Bones	30
		Shredded Napa Cabbage	15					De-stemmed Shiitake Mushrooms	8		Enoki Mushrooms	10		White Radish (Daikon) Chunks	40
		Sweet Potato Glass Noodles, 500 g/pack	3		Chilled Fresh Lean Pork Shreds	10		Chilled Fresh Minced Pork	10		Egg (cleaned)	10			
		De-stemmed Shiitake Mushrooms	10								Cilantro (trimmed)	1		Garlic	2
Seasonings	**Seasonings**	Finely Chopped Scallion	1	**Seasonings**	Finely Chopped Scallion	2	**Seasonings**	Finely Chopped Scallion	1	**Seasonings**	Finely Chopped Scallion	1.5	**Seasonings**	Finely Chopped Scallion	1
		Mature Ginger	1.5		Mature Ginger	1.5		Mature Ginger	1.5		Mature Ginger	1.5		Mature Ginger	1.5
		Garlic Cloves	1		Garlic Cloves	1		Garlic Cloves	1		Garlic Cloves	1.5		Garlic Cloves	1

5 Discussion and Conclusion

This project conducted multi-faceted research and practice around food–medicine-homology recipes in cafeterias for myopia prevention among Jiangsu students. The comprehensive discussion and conclusions are as follows:

(1) Through literature research, we clarified the dilemmas in myopia prevention and control and the link between diet and myopia. Through questionnaire surveys, we studied students' dietary status and needs, and then designed safe, student-recognized food–medicine-homology recipes. The promotion activities at multiple schools achieved remarkable results, and cooperation with Sunan Food Material Company successfully improved recipes. We innovatively established design principles featuring specific nutrients and food–medicine-homology ingredients, and conducted empirical analysis linking recipes with vision data, fully proving their feasibility and effectiveness.

(2) Compared with other myopia prevention methods—such as simply increasing outdoor activities (limited by time), eye drops (side effects), and eye exercises (requiring active student adherence)—this project's food–medicine-homology recipes rely on daily school cafeteria meals, allowing students to naturally prevent myopia while eating, making it more convenient and sustainable. It not only provides healthy dietary choices for students but also spreads traditional Chinese dietary therapy culture and enhances students' health awareness and dietary literacy, which is of great significance for reducing myopia rates among students in Jiangsu and even across the country.

3) In the future, the scope of recipe promotion can be further expanded to more schools, continuously tracking students' vision changes and dietary feedback, and constantly optimizing recipes. At the same time, cooperation with all parties should be strengthened, the deep relationship between ingredient nutrients and myopia prevention should be further studied, and the application system of traditional Chinese dietary therapy in myopia prevention and control should be improved to provide stronger protection for students' visual health.

References

[1] Lim, L. S., Gazzard, G., Low, Y. L., Choo, R., Tan, D. T., Tong, L., ... Saw, S. M. (2010). Dietary factors, myopia, and axial dimensions in children. Ophthalmology, 117(5), 993–997.

[2] Yin, C. J. (2022). Nutrients and myopia. Journal of Hygiene Research, 51(5), 720–724.

[3] Kim, J. M., & Choi, Y. J. (2024). Association between dietary nutrient intake and prevalence of myopia in Korean adolescents: Evidence from the 7th Korea National Health and Nutrition Examination Survey. Frontiers in Pediatrics, 11, Article 1285465. https://doi.org/10.3389/fped.2023.1285465

[4] Gardiner, P. A. (1960). Protein and myopia. Proceedings of the Nutrition Society, 19(1), 96–100.

[5] Ren, Z., Xu, P., Zhang, Q., Xu, J., Yang, Z., Si, X., et al. (2022). Relationship between sugary food intake and myopia in 11–14-year-old Chinese children in 2019–2021. Journal of Hygiene Research, 51(5), 713–719.

[6] Masenya, A. O. (2022). Role of nutrition on myopia progression [Thesis, School of Advanced Education, Research and Accreditation].

[7] Yao, F. W., & Zheng, Z. T. (1991). Dietary therapy for eye diseases Shanghai Scientific and Technical Publishers.

[8] Fan, L. L. (2011). A complete collection of medicinal diets by famous physicians . Zhejiang Science and Technology Publishing House.

[9] Ministry of Education of the People’s Republic of China and seven other departments. (2018, August 30). Notice on issuing the implementation plan for comprehensive prevention and control of myopia in children and adolescents [Policy document]. Ministry of Education of the People’s Republic of China. http://www.moe.gov.cn/srcsite/A17/moe_943/s3285/201808/t20180830_346672.html

[10] General Office of the National Health Commission of the People's Republic of China. (2023, July 25). Notice on issuing the ten core knowledge points for the prevention and control of myopia in children and adolescents [Policy document]. National Health Commission of the People's Republic of China. http://www.nhc.gov.cn/fys/s7906/202307/7a37c015f42340cf9387be594662bb73.shtml

[11] Cai, R. M. (2024). School-based research on the causes of and countermeasures for myopia in primary school students . Journal of the Chinese Society of Education, (S2), 1–2.

[12] Yang, R., Luo, X. X., & Zheng, W. R. (2024). Design of a clinical case report form for eye exercises in the treatment of myopia in children and adolescents . Chinese Journal of Chinese Ophthalmology, 34(7), 610–613.

[13] Duan, B. H. (2021). Design and compilation of recipes for vision protection in school-age children . China Food Safety Magazine, (10), 129–133.

[14] Shanghai Municipal Education Commission. (2024, October 8). Reply to Proposal No. 1148 of the Second Session of the 16th Shanghai Municipal People's Congress [Policy document]. Shanghai Municipal Education Commission. https://edu.sh.gov.cn/xxgk2_zhzw_jyta_02/20241008/ab016ff3c6694158ba195df496cd0c2e.html

[15] Wuzhou Municipal Education Bureau. (2024, September 3). "My recipe, my choice": The Municipal Education Bureau launches a nutritious recipe solicitation activity [Notice]. Wuzhou Education WeChat Account. https://mp.weixin.qq.com/s?__biz=MzIwODc1NDUwMQ==&mid=2247588277&idx=1&sn=be2056f40fc22e7c9742c618a067c7bc&chksm=9615da3d999d4c0a4e0da1c0391bdf84e4bb5b4813323a62413ffbac79aad0e1f0a63fb5464e&scene=27

[16] Holden, B. A., Fricke, T. R., Wilson, D. A., Jong, M., Naidoo, K. S., Sankaridurg, P., Wong, T. Y., Naduvilath, T. J., & Resnikoff, S. (2016). Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. Ophthalmology, 123(5), 1036–1042. https://pubmed.ncbi.nlm.nih.gov/26875007/

Appendix 1: First Questionnaire Survey

Hello! We are first-year high school students from the Jiangsu EaglePlate team, researching methods to prevent myopia.

The overall myopia rate among Chinese children and adolescents is 52.7%, the highest in the world. We would like to understand your views on myopia prevention and "food–medicine-homology" foods through a questionnaire.“Food–medicine homology” means that certain foods can serve both as daily food and as medicine. In our lives, some ingredients are not only delicious but can also have a positive impact on vision.

This questionnaire is for research purposes only, and information will be strictly confidential. You can fill it out with confidence. Every answer and opinion you give will help us protect children’s vision. Thank you for your support!

1. Which city in Jiangsu are you in?

2. What school level are you or your child in?

Kindergarten / Lower primary (grades 1–2) / Upper primary (grades 3–6) / Junior high school / Senior high school / Other

3. Do you or your child suffer from myopia?

Myopic and troubled / Myopic but not troubled / Not myopic / Unsure

4. Do you understand the concept of "food–medicine homology"—that some natural foods can be eaten and also treat illness? Understand / Do not understand

5. Do you think eating foods with medicinal value can help prevent myopia? Helpful / Not helpful / Unsure

6. Do you know that some foods, if eaten too much, may increase the risk of myopia? Know / Do not know

7. Which foods do you think may cause myopia if eaten often? (Multiple choice)

Cakes, milk tea, and other high-sugar foods / Fried foods, fatty meat, and other high-fat foods / Ham, sausage, salted meat, and other high-salt foods / Chili, garlic, pepper, and other stimulating foods / Unsure

8. Do you know that some foods, if eaten often, can help prevent myopia? Know / Do not know

9. Which of the following ingredients do you eat least in the cafeteria?

Vitamin A: pumpkin, banana, citrus, etc. / Vitamin B: animal liver, millet, etc. / Vitamin C: kiwi, tomato, strawberry, etc. / Lutein: carrot, broccoli, spinach, corn, etc. / Anthocyanins: blueberry, black goji berry, etc. / Protein: beef, fish, shrimp, eggs, beans, etc. / Other

10. Does the cafeteria provide food that can prevent myopia (such as food rich in vitamins, lutein, anthocyanins, or protein)? Often / Sometimes / Rarely / Never / Unsure

11. Are you satisfied with your school cafeteria's dishes?

Very satisfied / Somewhat satisfied / Average / Somewhat dissatisfied / Very dissatisfied

12. Do you know the most commonly used methods to prevent myopia currently? (Choose 3)

Use electronic products less / Do eye exercises / Do more outdoor exercise / Get enough sleep / Correct reading and writing posture / Take vision health supplements / Eat food–medicine-homology dishes / Other: [please specify]

13. What benefits do you think myopia-preventing food–medicine-homology dishes in the school cafeteria could offer?

Children like it more and can stick with it / No effort and no time / Less money / Can provide personalized prevention plans / Other: [please specify]

14. For myopia-preventing food–medicine-homology recipes, which expert's recommendation would you most like to hear?

Western medicine ophthalmologist / Traditional Chinese medicine ophthalmologist / Nutritionist / Chef / Education expert / Other: [please specify]

15. If the cafeteria provides myopia-preventing food–medicine-homology dishes, are you willing to buy them?

Very willing / Somewhat willing / Indifferent / Somewhat unwilling / Very unwilling

16. What ingredients would you prefer in myopia-preventing dishes? (Choose 3)

Fruits (such as blueberry, orange, etc.) / Vegetables (such as carrot, spinach, etc.) / Nuts (such as walnut, almond, etc.) / Grains

(such as corn, millet, etc.) / Chinese medicinal herbs (such as goji berry, chrysanthemums) / Meat and eggs (such as beef, fish, shrimp, etc.) / Other: [please specify]

17. If you want to buy myopia-preventing food–medicine-homology dishes, what would affect your decision? (Choose 3)

Reasonable price / Fresh ingredients / Delicious taste / Safe and hygienic / Good myopia prevention effect / Other: [please specify]

18. How much are you willing to pay per meal for myopia-preventing dishes?

Under 10 yuan / 10–20 yuan / 20–30 yuan / 30–40 yuan / Over 40 yuan

19. Do you think myopia-preventing dish recommendations should include nutritional knowledge?

Very necessary, hope for detailed nutritional knowledge / Necessary, some simple explanations are enough / Not necessary / Indifferent

20. How would you like to learn about myopia-preventing dishes in the school cafeteria?

Cafeteria tasting / Knowledge lectures / Posters, brochures / Douyin, Xiaohongshu, and other apps / Other: [please specify]

Appendix 2: Second Questionnaire Survey

Dear parents/students:

Hello! We are first-year high school students from the Jiangsu EaglePlate team, participating in the CTB competition. Given the severe situation that China's adolescent myopia rate is the highest in the world, we have launched the project "Design of Food–Medicine-Homology Recipes in Cafeterias for Myopia Prevention" and have conducted the first questionnaire survey to gain a deeper understanding. Traditional Chinese medicine has a long history and rich experience in dietary therapy, emphasizing "food–medicine homology"—believing that food can not only nourish the body but also prevent and treat diseases. We have deeply studied 56 dietary therapy books and 21 papers specifically for myopia prevention, consulted experts in Chinese and Western medicine, nutritionists, professors, and chefs, designed food–medicine-homology recipes for myopia prevention in cafeterias, and conducted publicity and promotion. To better optimize recipe design, we are now conducting the second questionnaire survey. Your opinions are very important for improving children's vision and cultivating scientific eating habits. Questionnaire information will be kept strictly confidential. Please fill it out with confidence. Thank you!

1. What type of school are you or your child in? (Single choice)

A. Kindergarten

B. Primary school

C. Junior high school

D. Senior high school E. University F. Other (please specify)

2. Do you or your child suffer from myopia? (Single choice)

A. Myopic and troubled

B. Myopic but not troubled

C. Not myopic

D. Unsure

3. Did you know the "food–medicine homology" concept before? (Single choice)

A. Yes

B. No

4. Do you understand the myopia-preventing effects of many natural ingredients? (Single choice)

A. Very knowledgeable

B. Somewhat knowledgeable

C. Know a little

D. Not very knowledgeable E. Not at all knowledgeable

5. If the school cafeteria provides the following myopia-preventing [food–medicine-homology dishes] long-term, which would you be willing to buy? (Choose 3 if willing)

A. Stir-fried pork liver with goji berries	B. Steamed young chicken with longan and goji berries	C. Goji berry and pig eye casserole	D. Astragalus beef
E. Stir-fried pork heart with goji leaves	F. Cold dressed celery	G. Boiled eggs with plantain seed	H. Carrot-stewed crucian carp
I. Stewed hen with mulberry	J. Stir-fried pork with paper	K. Astragalus and	L. Astragalus stewed

and white chrysanthemum	mulberry fruit, dodder seed, and daylily	angelica stewed beef	hen
M. Stewed lean pork with glossy privet fruit	N. Stewed lean pork with white chrysanthemum	O. Boiled river fish with goji berries	P. Stir-fried lean pork with goji berries
Q. Stir-fried lean pork with cassia seed	R. Stewed lean pork with pipewort	S. Stewed eggs with goji berries	T. Dodder seed eggs
U. Angelica sheep liver	V. Steamed chicken with astragalus and ginkgo	W. Radish and goji berry stewed duck liver	X. Steamed sea snail
Y. Stir-fried daylily with wood ear	Z. Smoked liver	AA. Stir-fried eel shreds with carrot	AB. Egg-sliding bitter melon slices
AC. Stewed bullfrog with goji berry and fish maw	AD. Braised pork kidney with goji berries	AF. Do not want to buy	

6. If the school cafeteria provides the following myopia-preventing [food–medicine-homology porridges] long-term, which would you be willing to buy? (Choose 3 if willing)

A. Goji berry porridge	B. Sour jujube seed porridge	C. Eight-treasure porridge with beans and japonica rice	D. Dodder seed porridge
E. Sheep liver porridge	F. Purslane and pork liver porridge	G. Sweet potato porridge	H. Plantain seed porridge
I. Yam and abalone shell porridge	J. Rehmannia and goji berry porridge	K. Longan and codonopsis porridge	L. Codonopsis and japonica rice porridge
M. Black rice, black bean, and sheep liver porridge	N. Black chicken porridge	O. Goji berry, chrysanthemum, and lean pork porridge	P. Radish, goji berry, and corn porridge
Q. Red date and chrysanthemum porridge	R. Black rice health porridge	S. Shepherd's purse porridge	T. Brown rice seafood egg yolk porridge
U. Do not want to buy			

7. If the school cafeteria provides the following myopia-preventing [food–medicine-homology soups] long-term, which would you be willing to buy? (Choose 3 if willing)

A. Sheep liver and goji berry soup	B. Kelp and cassia seed soup	C. Chicken liver and egg soup	D. Pipewort liver-nourishing soup
E. Scallion, pork liver, and egg soup	F. Fish eye and kelp soup	G. Spinach and pork liver soup	H. Goji leaf egg drop soup
I. Pork liver and spinach soup	J. Oyster, mushroom, and seaweed soup	K. Goji berry, chenpi, and red date soup	L. Goji berry, white fungus, jasmine, and chicken liver soup
M. Pork rib medicinal soup	N. Goji berry and pork liver soup	O. Pork liver, scallion, and egg soup	P. Goji berry, red date, and egg soup

Q. Pork liver, mulberry, goji berry, and double leaf vision-improving soup	R. Apple and snakehead fish soup	S. Pipewort and sheep liver soup	T. Beef bone, goji berry, and carrot soup
U. Carrot and corn pork bone soup	V. Sheep bone soup	W. Green vegetable and pork liver soup	X. Fresh milk, mushroom, and spinach thick soup
Y. Do not want to buy			

8. If the school cafeteria provides the following myopia-preventing [food–medicine-homology beverages] long-term, which would you be willing to buy? (Choose 3 if willing)

A. Chrysanthemum tea	B. Ginseng and goji berry drink	C. Goji berry, chenpi, and longan drink	D. Goji berry, chenpi, and red date drink
E. Sesame, walnut, and honey milk drink	F. Honey soy milk	G. Honey black sesame milk	H. Honey black sesame soy milk
I. Honey milk egg	J. Black sesame walnut milk	K. Jujube carrot drink	L. Pumpkin soy milk
M. Goji berry and chrysanthemum tea	N. Do not want to buy		

9. If the school cafeteria provides the following myopia-preventing [food–medicine-homology thick soups] long-term, which would you be willing to buy? (Choose 1 if willing)

A. Pork liver and egg thick soup

B. Chicken liver thick soup

C. Egg and milk thick soup

D. Walnut, jujube, and goji berry egg thick soup E. Pork liver thick soup F. Do not want to buy

10. Which type of myopia-preventing food would you or your child prefer the cafeteria to promote? (Single choice)

A. Dishes

B. Porridge

C. Soup

D. Beverages E. Thick soups F. Dried fruits G. Pastries H. Do not want to buy

11. Do you think the design of these recipes is scientific? (Single choice)

A. Very scientific

B. Somewhat scientific

C. Average

D. Not very scientific E. Not at all scientific

12. What price positioning for myopia-preventing food can you accept? (Single choice)

A. Equal to current cafeteria prices

B. Can be slightly higher than current cafeteria prices

C. Can be significantly higher than current cafeteria prices

D. Price is not the main consideration; the key is effectiveness

13. What measures do you think the school cafeteria can take to attract people to try myopia-preventing recipes? (Single choice)

A. Provide free tasting events

B. Hold related health lectures

C. Promote on social media

D. Cooperate with parents for joint promotion E. Other (please specify)

Thank you again for taking the time to fill out this questionnaire! If you are interested in our project or have any questions, please feel free to contact us.

Contact: [Email] eagleplate@126.com; [QQ Group] 2154026234. Thank you!

Appendix 3: Third Questionnaire Survey and Drink Flyer

In recent years, the global myopia rate among children and adolescents has continued to climb. The myopia rate among Chinese adolescents is the highest in the world, and Jiangsu students' myopia rate ranks among the highest of all provinces. Myopia has become a major public health challenge, and there is an urgent need to explore new effective prevention strategies. Chinese myopia dietary therapy culture has a long history. Our ancestors accumulated rich dietary therapy experience in the long struggle against eye diseases, and this valuable experience provides us with new ideas and methods.

We are the EaglePlate team participating in the CTB competition. Our research topic is exactly the design and promotion of food–medicine-homology recipes in cafeterias for myopia prevention—that is, preventing and alleviating myopia through dietary conditioning. We aim to deeply explore and inherit Chinese myopia dietary therapy culture, combine it with modern nutrition principles, and create recipes that are both delicious and have myopia-preventing effects, so that more students can understand and love this scientific way of eating, cultivate good eating habits, and effectively reduce myopia risk.

The two drinks you see now are the myopia-preventing food–medicine-homology drinks we have carefully prepared after consulting a large number of dietary therapy books. They have been reviewed one by one by Dr. Huang Xiuxian, a dietary therapy expert from the Traditional Chinese Medicine Department of Zhenjiang Integrated Traditional Chinese and Western Medicine Hospital, and Dr. Sun Dewei, Vice Dean of the School of Tourism and Culinary Arts at Wuxi Vocational Institute of Commerce. We sincerely invite you to taste them. If you have more suggestions or needs, please contact us in time. Our contact is [QQ Group] 2154026234.

Honey Black Sesame Milk Hot Drink

Honey occupies an important position in Chinese culture. In ancient times, there were specialized institutions for collecting honey as tribute. Honey wine was already served at Western Zhou court banquets. The Southern Song Dynasty's Lu You, in *Notes from the Old Xue An*, recorded that Su Dongpo and the monk Zhongshu became friends through their shared love of honey. Zhongshu liked to dip vegetarian dishes in honey, which others found hard to accept, but Su Shi was quite interested and even wrote "Song of the Old Man of Anzhou Eating Honey" for Zhongshu to express his love and appreciation for honey.

The Divine Farmer's Materia Medica even lists honey as a superior medicine, saying it "calms the five viscera and all deficiencies, supplements qi and centers, relieves pain and detoxifies, removes all pains, and harmonizes all medicines." About 70% of honey's sugars are glucose and fructose. These natural sugars are easily absorbed by the body, can quickly provide energy to eye tissues, and promote eye blood circulation, and ensure the eyes receive sufficient oxygen and nutrients. Honey is also rich in flavonoids, phenolic acids, and other antioxidants, which can quickly neutralize free radicals, reduce oxidative stress damage to the eyes, and provide strong protection for eye health.

When it comes to health-preserving foods, black sesame is also indispensable. Tao Hongjing, a pharmacologist of the Southern Dynasties, once evaluated sesame: "Among the eight grains, only this is excellent. Immortals make rice cakes with it and eat it, cutting off grains to attain longevity." The Compendium *of Materia Medica* states: "Taking black sesame for a hundred days can eliminate all chronic diseases. After one year, the body is glossy and not hungry; after two years, white hair turns black again; after three years, lost teeth grow back." Black sesame is rich in protein, fat, vitamin E, and other nutrients. Vitamin E and unsaturated fatty acids can resist oxidation and protect eyes from free radical damage; iron helps improve

blood circulation and provides sufficient oxygen and nutrients to the eyes. Long-term moderate consumption of black sesame can relieve eye fatigue and prevent dry eye and other diseases.

Milk, as a standout among daily drinks, pairs perfectly with honey and black sesame. The Song Dynasty poet Lu You, in his poem "Early Summer," once depicted: "Locust and willow form shade, rain washes the dust; cherries and dairy products are tasted fresh together." Milk is rich in vitamins A and D, which are extremely important for eye health. Vitamin A protects the cornea, and vitamin D helps reduce vision degeneration caused by aging. In addition, calcium in milk also has a positive effect on nerve transmission and signal function within the eye, further enhancing eye health and vitality.

Preparation: Prepare 1000 ml milk, 50 g roasted black sesame, 50 g honey. First, add 1000 ml milk to a blender or soy milk maker, then add 50 g ground roasted black sesame and 50 g honey. Use the soy milk mode to blend and heat, and it is ready to drink. The food–medicine-homology ingredients include black sesame and honey.

Goji Berry, Orange Peel, Red Date, and Honey Drink

The famous Southern Song poet Lu You loved reading and studied by lamplight for long periods, which unfortunately damaged his vision. However, in his later years, he accidentally discovered that his vision improved significantly after taking goji berries. To commemorate this experience, Lu You wrote "Book Matters at Yujian Studio," and the line "Morning studio, a cup of goji berry broth" not only reveals his habit of taking goji berry broth every morning to maintain his eyes but also witnesses the benefits of goji berries for vision.

Goji berries, chenpi, and red dates are all valuable food–medicine-homology ingredients. From the perspective of traditional Chinese medicine, goji berries nourish the liver and kidneys, benefit essence, and brighten the eyes; red dates supplement qi and nourish blood, benefiting eye protection; chenpi regulates qi and strengthens the spleen, dries dampness, and resolves phlegm. The three work synergistically to nourish the liver and kidneys, supplement the center and boost qi, regulate qi, and strengthen the spleen, providing sufficient nutrients for the eyes. From the perspective of modern nutrition, this drink is rich in various eye-beneficial nutrients. Among them, vitamins A and C are indispensable nutrients for eye health and can protect eyes from oxidative damage; goji berry polysaccharides have unique biological activity and positively promote eye health; trace elements such as iron, calcium, and phosphorus are also necessary for normal eye metabolism; in addition, flavonoids and volatile oils can also help restore vision and prevent myopia and other eye problems. Honey added to the drink not only adds sweetness but is also rich in various antioxidants and nutrients, complementing goji berries, chenpi, and red dates.

Preparation: Prepare 40 g goji berries, 100 g jujubes, 20 g chenpi, 50 g honey, and 2000 ml purified water in total (two batches). First, add 1000 ml of purified water to a pot, then add 40 g of goji berries, 100 g of jujubes (washed and cut open), 20 g of chenpi (washed), and 50 g of honey in sequence. After bringing to a boil over high heat, simmer over low heat for 20 minutes; strain the decoction and set aside. Add another 1000 ml of purified water to the pot, bring to a boil, then simmer for 20 minutes; strain and pour into the first batch of decoction, stir well, and it is ready to drink. The food–medicine-homology ingredients include goji berries, jujubes, chenpi, and honey.

Please leave your valuable feedback. Thank you very much! 1. What do you think the food–medicine-homology drink tastes like? A. Good B. Not good C. Did not drink 2. Do you think food–medicine-homology food has myopia-preventing effects? A. Yes B. No C. Do not know 3. Do you hope the school adds food–medicine-homology food? A. Add B. Do not add 4. Your suggestions for improving myopia-preventing food–medicine-homology food are:	(Please fill in)

Promoting food–medicine-homology recipes, protecting the campus "sunny" world—2024 China Thinks Big Innovation Research

Competition EaglePlate Team

Appendix 4: Three Categories of Myopia-Preventing Cafeteria Food–Medicine-Homology Recipes, 84 Dishes (2nd Edition)

(Including dishes, porridges, soups, beverages, dried fruits, pastries, etc.)

I. Nutritional Supplementation: Adequate Intake of Vitamin A, Riboflavin, and Other Nutrients to Maintain Normal Eye Function

1. [Stir-Fried Pork Shreds with Goji Berries] Food–medicine-homology ingredient: goji berries.

Ingredients: 50 g goji berries, 200 g lean pork, 10 g lettuce, appropriate amount of cooking seasonings.

Method: Cut lean pork and lettuce into thin shreds. When the oil is 70% hot, stir-fry meat shreds and lettuce shreds. Add cooking wine, soy sauce, salt, and MSG, then add goji berries and stir-fry.

Effects: Goji berries nourish the liver and kidneys, moisten the lungs, and brighten the eyes. Goji berries contain a high amount of carotene, which can be converted into vitamin A in the body to maintain normal vision. Pork is rich in protein; by tonifying the body, it makes qi and blood flourish to nourish eye tissues.

2. [Stir-Fried Pork Liver with Goji Berries] Food–medicine-homology ingredient: goji berries.

Ingredients: 50 g goji berries, 250 g pork liver, appropriate amounts of lard, salt, cooking wine, and MSG.

Method: Soak goji berries in warm water for 2 hours and remove. Slice pork liver, mix with salt and cooking wine, stir-fry in lard until nearly cooked, add goji berries and stir-fry until cooked. Add MSG and serve. Eat with meals in portions.

Effects: Nourishes the liver and kidneys, nourishes blood, and brightens the eyes. Indicated for symptoms such as difficulty seeing near objects, gradually appearing black spots before the eyes, dizziness, tinnitus, and soreness and weakness of the waist and knees.

3. [Steamed Young Chicken with Longan and Goji Berries] Food–medicine-homology ingredients: longan flesh, goji berries, jujubes. Ingredients: 1 young chicken, 30 g each of longan flesh, goji berries, and red dates.

Method: Remove feathers and internal organs from the young chicken and wash. Put longan flesh, goji berries, and red dates into the chicken cavity. Steam until cooked and season to taste.

Effects: Nourishes blood and strengthens the spleen, nourishes the liver, and brightens the eyes. Indicated for dry eyes, dizziness, palpitations, insomnia, fatigue, and soreness and weakness of the waist and knees.

4. [Stir-Fried Eel Shreds with Carrot]

Ingredients: 200 g eel, 2 carrots.

Seasonings: appropriate amounts of vegetable oil, salt, soy sauce, and vinegar.

Method: Wash eel and cut into thin shreds; remove roots from carrots, wash, and shred. Heat vegetable oil in a pan, add eel shreds and carrot shreds, stir-fry, add salt, soy sauce, and vinegar, stir-fry until cooked, and serve.

Effects: Carrots protect the eyes and can prevent and treat night blindness and dry eye disease. Eel helps maintain normal vision and prevent eye diseases. Regular consumption of this dish has a good effect on eye health.

5. [Steamed Sea Snail] Food–medicine-homology ingredient: ginger.

Ingredients: 350 g sea snail meat, a few ginger chunks and scallion sections.

Seasonings: appropriate amounts of cooking wine, salt, MSG, soy sauce, vinegar, Sichuan pepper, and sesame oil.

Method: (1) Rub and wash sea snail meat repeatedly, cut into two connected pieces, flatten with a knife, score diagonal cuts on both sides, and cut open from the middle; half of the ginger minced, half crushed. Mix soy sauce, vinegar, salt, and minced ginger into a ginger sauce.

(2) Place sea snail meat on a plate, add cooking wine, MSG, scallion sections, crushed ginger, Sichuan pepper, and water. Steam for about 15 minutes, remove, pick out the ginger, scallion, and Sichuan pepper, drizzle with sesame oil, and serve with ginger sauce.

Effects: Sea snail meat is rich in vitamin A. Vitamin A helps synthesize rhodopsin in the retina, which can help relieve eye fatigue, dryness, and other symptoms, and has a certain auxiliary effect on myopia prevention.

6. [Smoked Liver Tips]

Ingredients: 350 g fresh pork liver, 50 g rehydrated black fungus, 1/2 cucumber, a little chopped scallion and garlic slices.

Seasonings: appropriate amounts of vegetable oil, soy sauce, salt, MSG, water starch and sesame oil.

Method: (1) Wash pork liver, cut into thin slices, mix with soy sauce and water starch; remove roots from black fungus, wash, and tear into pieces; wash cucumber and slice. (2) Heat vegetable oil in a pan to 50–60% hot, add coated pork liver slices, remove and drain oil. (3) Leave a little oil in the pan, sauté chopped scallions and garlic slices, add soy sauce, black fungus, cucumber slices, water, salt, and MSG, bring to a boil, skim off foam, and thicken with water starch. (4) Add pork liver, drizzle with sesame oil, stir a few times, and serve.

Effects: Pork liver is nutritious and has the functions of tonifying the liver, nourishing blood, and benefiting the eyes. It is mainly used for blurred vision, dry eyes, night blindness, and red eyes.

7. [Angelica Sheep Liver] Food–medicine-homology ingredients: angelica, ginger. This dish can also be classified as Category 2.

Ingredients: 10 g each of angelica and scallion white, 100 g sheep liver, 5 g each of ginger and garlic paste, 20 ml each of cooking wine and soy sauce, 3 g refined salt, 3 ml sesame oil, 2 g MSG.

Method: (1) Wash angelica, soften in warm water, slice, and put into a gauze bag; cut scallion into sections, slice ginger. (2) Wash the sheep liver, put it into a clay pot, add an appropriate amount of water, put in the medicine bag, scallion sections, ginger slices, cooking wine, and refined salt. Bring to a boil over high heat, skim off foam, switch to low heat, and cook until the sheep liver is thoroughly cooked. Remove. (3) Mix soy sauce, MSG, garlic paste, and sesame oil into a sauce. (4) Cut cooked sheep liver into thin slices, place on a plate, and drizzle with the prepared sauce.

Effects: Benefits the liver, nourishes the liver, brightens vision, and moistens hair. Used in Western medicine as an auxiliary treatment for premature graying of hair, hyperopia, myopia, cataracts, and glaucoma.

Contraindications: Those with excessive dampness and fullness in the middle, or diarrhea, should avoid angelica; those with high blood lipids and high cholesterol should not eat too much sheep liver.

8. [Radish and Goji Berry Stewed Duck Liver] Food–medicine-homology ingredients: goji berries, longan, ginger.

Ingredients: 250 g radish, 20 g goji berries, 150 g duck liver, 6 g each of scallion sections and ginger slices, 100 g lard, 6 ml cooking wine, appropriate amount of refined salt.

Method: (1) Wash and peel radish, shred, and cook; wash goji berries; wash duck liver, cut into thin slices with a flat knife, and blanch in boiling water. (2) Place a pan over medium heat, add lard and an appropriate amount of water, scallion sections, ginger slices, cooking wine, refined salt, radish shreds, and goji berries. Switch to high heat and stew until the juice is thick, then add duck liver, stir-fry until cooked, and remove from heat.

Effects: Eat meat and drink soup, either alone or with meals, finishing in 1–2 servings. Has the effect of clearing the liver and brightening the eyes, rich in vitamin A. Suitable for those with dry eyes, excessive tears, blurred vision, and decreased vision.

9. [Boiled River Fish with Goji Berries] Food–medicine-homology ingredient: goji berries.

Ingredients: 50 g goji berries, 1 river fish (about 250 g).

Method: Remove impurities from river fish, wash, and cook in soup with goji berries, wine, salt, and other seasonings.

Effects: Eat with lunch and dinner. Suitable for myopia patients.

10. [Carrot Stewed Crucian Carp]

Ingredients: 3–5 carrots, 1 fresh crucian carp.

Method: Remove scales and internal organs from the fresh crucian carp, wash, and stew with carrots in clear water.

Effects: Suitable for myopia patients.

11. [Goji Berry and Pig Eye Casserole] Food–medicine-homology ingredients: goji berries, ginger. This dish can also be classified as Category 2.

Ingredients: 30 g goji berries, 2 pairs pig eyes, 100 g longan flesh, 15 g Cornus officinalis.

Method: Wash goji berries and Cornus officinalis, wrap in cloth; wash pig eyes. Heat a little lard in a pan, fry ginger and garlic until golden, add pig eyes, longan, cooking wine, and refined salt, add water, and bring to a boil. Skim off foam, add medicine bag, and after the soup boils, pour into a clay pot. Stew pig eyes until soft and tender, sprinkle with chopped scallion and chicken essence.

Effects: Pig eyes are crispy and tender, can nourish the liver and kidneys, benefit essence, and brighten the eyes. Suitable for those with liver and kidney deficiency and decreased vision. Adolescents with myopia and other patients eating it can help with myopia recovery.

12. [Goji Berry Seafood Chicken Soup Casserole] Food–medicine-homology ingredient: goji berries.

Ingredients: 50 g each of rehydrated sea cucumber, fresh clam meat, fresh mussel meat, cooked sea snail meat, and goji berries; 20 g each of rehydrated dried scallop and rehydrated abalone shreds; 1 abalone shell; appropriate amounts of chicken soup, MSG, yellow wine, and bamboo shoot slices.

Method: Put all into a clay pot, add chicken soup, simmer over low heat until 90% cooked, add bamboo shoot slices, yellow wine, and MSG, cook until done, and eat.

Effects: Suitable for myopia patients.

13. [Goji Berry Eggs] Food–medicine-homology ingredient: goji berries.

Ingredients: 2 eggs, 30 g goji berries.

Method: Boil eggs and goji berries in an appropriate amount of water together. After eggs are cooked, remove shells and cook for a while longer.

Effects: Eat eggs and drink soup for 3–5 consecutive days. Suitable for myopia.

14. [Sheep Liver and Goji Berry Soup] Food–medicine-homology ingredients: goji berries, ginger.

Ingredients: 100 g sheep liver, 50 g goji berries, appropriate amounts of minced ginger, salt, MSG, and other seasonings.

Method: Wash sheep liver, chop, stir-fry briefly in 70% hot oil, add an appropriate amount of water, add goji berries, simmer over low heat until the liver is cooked and soft, add seasonings.

Effects: Eat in the morning and evening. This is a folk dietary therapy recipe for eye health. It can be used for myopia caused by insufficient essence and blood of the liver and kidneys. Chicken liver or pork liver can also be used instead of sheep liver.

15. [Beef Bone, Goji Berry, and Carrot Soup] Food–medicine-homology ingredients: goji berries, ginger.

Ingredients: 250 g beef bones, 50 g goji berries, 150 g carrots.

Method: Crush beef bones, wash and cut carrots into pieces, wash goji berries, put all in a pot with an appropriate amount of water, simmer over low heat to fully dissolve marrow into the soup, add a little ginger, fish sauce, and MSG to season. Drink soup and eat goji berries and carrots.

Effects: Fills essence and benefits marrow, nourishes blood, promotes hair growth, brightens the eyes, and tonifies the liver. Used for hair that is prone to splitting, breaking, and falling out due to essence and blood deficiency. This recipe can be used as a home-cooked dish.

16. [Pork Liver, Mulberry, Goji Berry, and Double Leaf Vision-Improving Soup] Food–medicine-homology ingredient: mulberry leaf. This dish is also classified as Category 3.

Ingredients: 200 g pork liver, 150 g goji leaves, 40 g mulberry leaves (available in Chinese medicine stores), appropriate amounts of bean powder, refined salt, and oil.

Method: Wash pork liver, cut into thin slices, mix with bean powder, stir-fry in oil until cooked, remove and set aside; wash goji leaves and mulberry leaves, put in a pot, add water, bring to a boil over high heat, add the cooked pork liver, switch to low heat and simmer for half an hour, add refined salt to season. Eat with meals.

Effects: Has the effects of brightening the eyes, clearing heat, and quenching thirst. Commonly used in folk medicine to treat decreased vision, conjunctivitis, night blindness, etc.

17. [Chicken Liver and Egg Soup] Food–medicine-homology ingredient: ginger.

Ingredients: 150 g chicken liver, 2 eggs.

Method: Heat a little lard to 90% hot, fry ginger and garlic until golden, discard, add chicken liver and stir-fry briefly, add stock and bring to a boil, add eggs, refined salt, chopped scallion, and chicken essence.

Effects: Tonifies the liver and kidneys, nourishes blood, calms the fetus, calms the five viscera, and beautifies. Suitable for adolescents with myopia and patients with qi and blood deficiency and premature aging.

18. [Scallion, Pork Liver, and Egg Soup]

Ingredients: 150 g pork liver, 2 eggs, a little scallion white, appropriate amount of salt.

Method: Cook pork liver in water to make soup, add eggs and scallion white, season with salt, eat the pork liver, and drink the soup.

Effects: Tonifies deficiency, nourishes blood, and brightens the eyes. Used for myopia of liver blood deficiency type.

19. [Green Vegetable and Pork Liver Soup]

Ingredients: 200 g pork liver, 100 g green vegetables, 1 tbsp oil (about 15 ml), 1 tsp salt (about 5 g), 1/2 tsp chicken essence (about 3 g) 1 tbsp cooking wine (about 15 ml).

Method: Wash pork liver, cut into thin slices, marinate with cooking wine for a while, blanch and drain; wash green vegetables and drain. Heat oil in a pan, stir-fry pork liver until fragrant, add clean water, and bring to a boil over high heat. When the pork liver is cooked, add washed green vegetables and continue cooking for about 2 minutes. Add salt and chicken essence, and once flavored, serve.

Effects: Pork liver is sweet and warm in nature, with three major effects: tonifying the liver, nourishing blood, and benefiting the eyes. Pork liver stores fat-soluble vitamin A, with the highest content among daily foods. Vitamin A deficiency in the body affects vision; regular consumption of pork liver can increase corneal smoothness and luster, making eyes bright and spirited.

20. [Oyster, Mushroom, and Seaweed Soup] Food–medicine-homology ingredients: oyster, ginger.

Ingredients: 200 g fresh oyster meat, 50 g mushrooms, 30 g seaweed, appropriate amounts of ginger, sesame oil, salt, and MSG.

Method: First boil mushroom and ginger for 20 minutes, then add oyster and seaweed and cook briefly. Add seasonings and eat with soup. Effects: Has the effects of nourishing the kidneys and liver, nourishing blood, and brightening the eyes. Suitable for myopia patients.

21. [Apple and Snakehead Fish Soup] Food–medicine-homology ingredients: winter melon, ginger.

Ingredients: 3 apples, 1 fish, 10 red dates, 2 slices of ginger, a little salt.

Method: Peel, core, and stem apples, cut into pieces; remove pits from red dates. Clean fish, remove bones and spines, fry in a pan over medium heat until slightly yellow, remove. Add water to a clay pot, bring to a boil over high heat, add all ingredients, switch to medium heat, and simmer for about 2 hours. Add salt to season.

Effects: This soup can treat spleen deficiency and insufficient blood and qi, prevent eye bags, and eliminate dark circles under the eyes.

22. [Fish Eye and Kelp Soup] Food–medicine-homology ingredient: kelp.

Ingredients: 100 g each of fish eyes and fresh kelp, 15 g scallion, appropriate amounts of soy sauce and cooking wine.

Method: Wash fish eyes; soak kelp until soft, wash, and cut into 5 cm sections. Add water to a pot and bring to a boil. Add fish eyes and kelp. Cook over medium heat with the lid open. Add a little cooking wine midway to remove fishiness. Add chopped scallions and season with soy sauce.

Effects: Fish eyes are rich in DHA, which is an important substance for maintaining retinal and optic nerve function.

23. [Goji Berry, Chenpi, and Red Date Soup] Food–medicine-homology ingredients: goji berries, chenpi, jujubes, honey.

Ingredients: 20 g goji berries, 5 g chenpi, 10 red dates, 1 tbsp honey.

Method: Put goji berries, chenpi, and red dates in a pot, add an appropriate amount of water, bring to a boil over high heat, then simmer over low heat for 20 minutes. After cooling, add honey and drink.

Effects: Twice daily. Suitable for adolescents with pseudomyopia.

24. [Goji Berry, White Fungus, Jasmine, and Chicken Liver Soup] Food–medicine-homology ingredients: goji berries, ginger.

Ingredients: 100 g chicken liver, 15 g each of white fungus and goji berries, 20 jasmine flowers, appropriate amounts of water bean powder, cooking wine, ginger juice, clear soup, salt, and MSG.

Method: Wash chicken liver, slice, mix with water bean powder, cooking wine, ginger juice, and salt; wash white fungus, tear into small pieces, soak in clean water; remove flower bases from jasmine flowers, wash, and place on a plate; wash goji berries. Place a pot over heat, add clear soup, cooking wine, ginger juice, salt, MSG, white fungus, chicken liver, and goji berries. Bring to a boil, skim off foam, and after the chicken liver is cooked, put it into a bowl and sprinkle with jasmine flowers.

Effects: Eat with dinner. Suitable for myopia patients.

25. [Goji Berry and Pork Liver Soup] Food–medicine-homology ingredient: goji berries.

Ingredients: 100 g pork liver, 20 g goji berries.

Method: Cook soup and eat.

Effects: Suitable for myopia patients.

26. [Goji Berry, Red Date, and Egg Soup] Food–medicine-homology ingredients: goji berries, jujubes.

Ingredients: 15–30 g goji berries, 6–8 red dates, 2 eggs.

Method: Boil goji berries, red dates, and eggs in water together. After the eggs are cooked, remove the shells and simmer over low heat for half an hour.

Effects: Eat the eggs and drink the soup. Once daily or every other day. Suitable for myopia.

27. [Carrot and Corn Pork Bone Soup] Food–medicine-homology ingredient: ginger.

Ingredients: 1 pork bone, 1 corn, 1 carrot, appropriate amount of salt, 1 scallion, 4 slices of ginger, 2 sprigs of cilantro.

Method: Cut pork bone into pieces. First, soak in cold water for two hours to remove some blood. Add an appropriate amount of water to a pot, add ginger slices, and scallion sections. After the water boils, add pork bone; after the water boils again, add a large spoonful of cooking wine. Put the blanched pork bone in clean water and wash off the foam. Put cleaned pork bones in a soup pot, add a few sections of scallion and ginger slices again. Bring to a boil over high heat, then simmer over medium-low heat for 30 minutes. Add corn pieces and carrot pieces, switch to low heat, and continue simmering for one hour. Add an appropriate amount of salt half an hour before serving, and add a little chopped cilantro after turning off the heat.

Effects: Corn contains zeaxanthin, which has good eye-protecting effects; carotene in carrots can be converted into vitamin A in the body, also a good eye protector.

28. [Sheep Bone Soup] Food–medicine-homology ingredient: ginger.

Ingredients: 2 jin (about 1 kg) sheep bones, 1 piece of sliced ginger, scallion sections, appropriate amounts of star anise, cinnamon, galangal, and bay leaves.

Method: (1) Blanch sheep bones to remove gamey smell. Put sheep bones in cold water, bring to a boil over high heat, skim off foam, remove sheep bones, rinse with clean water, and set aside. (2) Heat oil in a pan. After heating, pour out the hot oil and add cold oil. Add ginger slices and scallion sections and stir-fry until fragrant. Add prepared spices and sheep bones. Stir-fry over medium heat for 3 minutes to release the spice aroma and moisture from the sheep bones. After smelling the meat's fragrance, drizzle half a spoonful of high-proof white wine, stir-fry to carry away the gamey smell through alcohol evaporation. After the alcohol completely evaporates, pour in an appropriate amount of boiling water to cover the sheep bones. After the soup boils again, skim off the foam, cover, and simmer over medium heat for 50 minutes. (3) Do not over-season the sheep bone soup. After 50 minutes, remove the spices from the pot, add a little salt and pepper to adjust the base flavor. Over-seasoning will only affect the freshness of the soup.

Effects: Treats consumptive disease, loss of essence, and decreased vision.

29. [Fresh Milk Mushroom Spinach Thick Soup]

Ingredients: 200 g tender spinach, 120 g potato, 1 small clove of garlic, an appropriate amount of fresh milk, 300 g button mushrooms, 1 small piece of butter, appropriate amounts of salt and black pepper, 1 spoonful of fresh cream, a handful of white sesame seeds.

Method: (1) Peel and cut the potato into small pieces, peel the garlic, and set aside. Wash the spinach and cut into small sections, wash the button mushrooms, and cut into thin slices. Boil potato pieces and garlic in water until cooked and soft, about 15 minutes after the water boils. Blanch spinach sections in boiling water, remove, rinse with cold water, and drain. (2) Put cooked potato, garlic, and drained spinach into a blender, add an appropriate amount of fresh milk (just covering the ingredients), and blend into a thick soup. After blending, heat slightly, and add a little salt to taste. (3) Heat a pan, add a small piece of butter to melt, add sliced button mushrooms, stir-fry until fragrant, add appropriate amounts of salt and black pepper to season. (4) Pour blended spinach thick soup into a large bowl, top with a spoonful of stir-fried button mushrooms, and finally sprinkle with a handful of white sesame seeds to enhance texture and appearance.

Effects: Spinach is rich in vitamins A, C, E, lutein, carotene, and other nutrients, all of which have good protective effects on the eyes. Among them, lutein can filter blue light, reduce ultraviolet damage to the eyes, and prevent age-related macular degeneration; carotene can be converted into vitamin A, promoting retinal development and function. In addition, mushrooms and fresh milk also contain eye-beneficial nutrients such as vitamin D and calcium.

30. [Pork Liver and Spinach Soup]

Ingredients: 100 g pork liver (sliced), 50 g spinach.

Method: Wash and cook together in soup. Add scallion, salt, and MSG to season.

Effects: Suitable for myopia patients.

31. [Radish, Goji Berry, and Corn Porridge] Food–medicine-homology ingredient: goji berries.

Ingredients: 250 g radish, 20 g goji berries, 10 g corn flour.

Method: Wash ingredients, mix corn flour with a little water into a paste, cut radish into small pieces, put into a pot with water. Bring to a boil over high heat, add goji berries and corn paste, stirring while adding. After boiling, switch to low heat and simmer until the corn paste is cooked and soft.

Effects: Take twice daily, morning and evening. Tonifies the kidneys, nourishes blood, brightens the eyes, and has the effects of promoting digestion, regulating qi, and relieving thirst. Radish is rich in carotene. Fresh radish juice is effective for diabetes; dripping juice into the nose (opposite side of migraine) is also effective. Regular consumption of radish porridge can keep you fit and protect your eyes.

32. [Black Rice Health Porridge] Food–medicine-homology ingredient: goji berries.

Ingredients: 100 g black rice, 20 g walnut kernels, 20 g goji berries, 100 g carrot.

Method: Wash and cut carrots into pieces. Wash black rice. Put a pot on heat, add an appropriate amount of water, add black rice, walnut kernels, and carrot pieces, bring to a boil over high heat, add goji berries, bring to a boil again, switch to low heat, and cook until the rice is done.

Effects: Benefits intelligence and the brain, nourishes blood, and brightens the eyes.

33. [Sour Jujube Seed Porridge] Food–medicine-homology ingredient: sour jujube seed.

Ingredients: 30 g sour jujube seed, 50 g japonica rice, 20 g brown sugar.

Method: Crush sour jujube seeds, wrap in gauze, put into a pot with japonica rice, add 600 ml water, cook until rice is soft and soup is thick, then turn off the heat. Remove sour jujube seeds, add brown sugar, mix well, cover, and let stand for about 5 minutes. Take warm 1 hour before bedtime.

Effects: Tonifies the heart and boosts qi, calms the mind, and stabilizes the will. Mainly treats symptoms such as clear near vision, blurred distance vision, pale complexion, palpitations, and fatigue.

34. [Goji Berry Porridge] Food–medicine-homology ingredient: goji berries.

Ingredients: 50 g goji berries, 100 g japonica rice.

Method: Cook goji berries and japonica rice with water into a thin porridge and eat.

Effects: Eat in the morning and evening. Goji berry porridge has the effects of nourishing the liver and kidneys and brightening the eyes. It is very suitable for myopia of liver and kidney yin deficiency type. Dodder seed can enhance the liver-nourishing and vision-brightening effect.

35. [Eight-Treasure Porridge with Beans and Japonica Rice] Food–medicine-homology ingredients: red beans, hyacinth beans, coix seed, longan flesh, lotus seeds, jujubes. This dish can also be classified as Category 2.

Ingredients: 20 g each of red beans, hyacinth beans, peanut kernels, coix seed, walnut kernels, longan flesh, lotus seeds, and red dates; 50 g japonica rice.

Method: Put all ingredients into a pot, add 700 ml of water, simmer over low heat to make porridge, and mix in a little brown sugar. Take warm.

Effects: Strengthens the spleen and boosts qi, benefits qi, brightens the complexion and the eyes. For symptoms such as blurred vision, inability to endure prolonged viewing, poor sleep and poor appetite, and indigestion.

36. [Codonopsis and Japonica Rice Porridge] Food–medicine-homology ingredient: codonopsis.

Ingredients: 100 g japonica rice, 20 g codonopsis.

Method: Decoct codonopsis twice in water, remove residue, add japonica rice and cook into porridge.

Effects: 1–2 times daily, for 3–4 consecutive weeks. Suitable for myopia patients.

37. [Sheep Liver Porridge]

Ingredients: 1 sheep liver, 20 g scallion seeds, 100 g japonica rice.

Method: Wash sheep liver, remove fascia, chop; stir-fry scallion seeds and grind into powder. Cook the two in a clay pot, take the juice, and cook porridge with japonica rice.

Effects: Especially suitable in winter. Warms and tonifies the liver and kidneys, brightens the eyes. Suitable for myopia of liver and kidney deficiency type.

38. [Black Rice, Black Bean, and Sheep Liver Porridge] Food–medicine-homology ingredient: ginger.

Ingredients: 60 g each of black rice, black beans, and sheep liver; appropriate amounts of salad oil, soy sauce, ginger shreds, and refined salt.

Method: Wash black rice and black beans separately, add an appropriate amount of water, and simmer into porridge. Wash sheep liver, chop, take a frying pan, add salad oil, soy sauce, ginger shreds, refined salt, etc., and stir-fry with sheep liver until cooked. Eat with black bean porridge.

Effects: Nourishes yin and tonifies the kidneys, brightens the eyes. Suitable for adolescent myopia patients.

39. [Brown Rice Seafood Egg Yolk Porridge] Food–medicine-homology ingredient: ginger.

Ingredients: appropriate amount of brown rice, appropriate amount of seafood (such as shrimp, crab stick, oyster, etc.), 2 salted egg yolks, appropriate amounts of ginger, lettuce, cooking oil, pepper, and salt.

Method: (1) Wash brown rice, soak for about 30 minutes to make it easier to cook. Heat oil in a pan, add sliced ginger and shrimp heads (if using fresh shrimp), and stir-fry over low heat to release shrimp oil. This step adds freshness to the porridge. (2) Remove the stir-fried shrimp heads and ginger slices, add an appropriate amount of boiling water, then add the soaked brown rice, bring to a boil over high heat, switch to low heat, cover, and cook for about 20 minutes until the brown rice is soft and tender. (3) Add crushed salted egg yolks to the porridge, stir well, and continue cooking for 2 minutes. Then add cut shrimp, crab sticks, oysters, and other seafood, and cook until the seafood changes color. (4) Add appropriate amounts of salt and pepper to season, add shredded lettuce, stir briefly, and serve.

Effects: Brown rice is rich in vitamin A and mineral elements, helping maintain normal eye physiological functions; seafood is rich in omega-3 fatty acids, beneficial to the retina; egg yolk is rich in vitamin A, phospholipids, vitamin B, and other nutrients, helping protect vision and prevent eye diseases.

40. [Goji Berry, Chenpi, and Red Date Drink] Food–medicine-homology ingredients: goji berries, chenpi, jujubes, honey.

Ingredients: 10 g goji berries, 3 g chenpi, 8 red dates, 2 tsp honey.

Method: Put goji berries, chenpi, and red dates in a pot, add an appropriate amount of water, simmer over low heat for 20 minutes, take the first decoction, add water and cook again, take the second decoction.

Effects: Mix first and second decoctions, divide into two doses, take 3–4 hours apart, add 1 tsp honey when taking. Goji berries are rich in carotene and vitamin A. Antioxidants such as anthocyanins in goji berries also help protect eyes from free radical damage. Chenpi has the effect of regulating qi and strengthening the spleen, helping improve spleen and stomach function and promoting digestion and absorption. Good spleen and stomach function helps the body absorb and utilize nutrients, thus positively affecting eye health. Red dates are rich in vitamin C and vitamin A; vitamin C helps protect eyes from free radical damage, while vitamin A is essential for maintaining normal visual function.

41. [Honey Black Sesame Milk] Food–medicine-homology ingredients: black sesame, honey.

Ingredients: honey, black sesame, milk.

Method: Appropriate amount of black sesame. Stir-fry until cooked, grind into fine powder, take 1 tbsp each time, mix into 1 cup of milk, add 1 tsp of honey, and take.

Effects: Suitable for myopia patients.

42. [Honey Black Sesame Soy Milk] Food–medicine-homology ingredients: black sesame, honey.

Ingredients: honey, black sesame, soy milk.

Method: Appropriate amount of black sesame. Stir-fry until cooked, grind into fine powder, take 1 tbsp each time, mix into 1 cup of soy milk, add 1 tsp of honey, and take.

Effects: Suitable for myopia patients.

43. [Honey Milk Egg] Food–medicine-homology ingredient: honey.

Ingredients: 100 ml milk, 1 tsp honey, 1 egg.

Method: Beat egg thoroughly, pour into heated milk, simmer over low heat until boiling. After the egg is cooked, add honey.

Effects: Can be eaten as a snack or with bread or steamed buns for breakfast. Eggs and milk are complete and balanced in nutrients. Honey also contains various nutrients; its sugars are mainly fructose, and eating a small amount will not raise blood sugar. These are all high-grade nutritious foods. Long-term consumption can maintain

eye and overall health, enhance disease resistance, and has preventive and therapeutic effects on various types of myopia.

44. [Jujube Carrot Drink] Food–medicine-homology ingredient: jujubes.

Ingredients: 5 carrots, 7 jujubes.

Method: Decoct in water and consume regularly.

Effects: Suitable for myopia patients.

45. [Pumpkin Soy Milk]

Ingredients: 100 g each of pumpkin and soybean.

Seasoning: appropriate amount of rock sugar.

Method: The night before, wash the pumpkin skin, keep the skin and seeds, cut into small pieces, and set aside. The night before, wash the soybeans, put them into a soy milk maker, and add an appropriate amount of water, and soak overnight. Put pumpkin pieces into the soy milk maker, add an appropriate amount of rock sugar according to taste. Connect the power, select the "soy milk" button, cook until the soy milk is done, and filter.

46. [Black Sesame Walnut Milk Honey Drink] Food–medicine-homology ingredients: black sesame, honey.

Ingredients: 250 g each of black sesame, walnut kernels, and milk; appropriate amount of honey.

Method: Stir-fry sesame until fragrant, grind into fine powder; slightly stir-fry walnut kernels and mash. Store separately in bottles. Take 1 tbsp each time, mix well, and take warm. 1–2 times daily.

Effects: Nourishes the liver and kidneys, benefits essence, nourishes blood, brightens the eyes, and moistens dryness. Mainly treats symptoms such as difficulty seeing near objects, gradually appearing black spots before the eyes, and dry eyes.

47. [Walnut, Jujube, and Goji Berry Egg Thick Soup] Food–medicine-homology ingredients: jujubes, goji berries.

Ingredients: 300 g walnut kernels, 250 g red dates (pitted), 150 g goji berries, 500 g fresh pork liver.

Method: Slightly stir-fry walnut kernels and remove the skin, chop together with the above ingredients, put into a porcelain bowl, add a little water, stew in a water bath for half an hour, and set aside. Each time, take 1 tbsp, crack in 1 egg, add a little brown sugar, steam into a thick soup, and take warm. Twice daily.

Effects: Benefits the kidneys and tonifies the liver, nourishes blood, and brightens the eyes. Mainly treats decreased vision, or accompanied by dizziness, forgetfulness, and soreness and weakness of the waist and knees.

48. [Egg and Milk Thick Soup] Food–medicine-homology ingredient: honey.

Ingredients: 1 egg, 1 cup milk, 1 tsp honey.

Method: Beat egg thoroughly, pour into heated milk, simmer over low heat until boiling. After the egg is cooked, let cool, then add honey.

Effects: Can be eaten after breakfast or as a morning snack, with bread or steamed buns. Eggs, milk, and honey can play a certain role in nourishing eye muscles, sclera, retina, optic nerve, and other tissues. Through nutrition, they can enhance the strength of the ciliary muscle and the toughness of the sclera.

49. [Pork Liver Thick Soup]

Ingredients: pork liver slices, fermented black beans, scallion white, egg, salt, MSG, etc.

Method: Add an appropriate amount of water to a pot, add 100 g pork liver slices, cook until the liver is done, add fermented black beans and scallion white, then crack in an egg, add salt and MSG to season.

Effects: Pork liver is rich in vitamin A, can nourish the eyeball, and has the effect of nourishing the liver and brightening the eyes.

50. [Pork Liver and Egg Thick Soup]

Ingredients: 100 g pork liver (sheep liver, beef liver, or chicken liver can also be used), 1 egg, appropriate amounts of fermented black beans, scallion white, salt, and MSG.

Method: Wash pork liver, cut into slices. Put in a pot with an appropriate amount of water, simmer over low heat until the liver is cooked, add fermented black beans and scallion white, then crack in an egg, add salt and MSG to season.

Effects: Eat with meals. Eggs and pork liver are both protein-rich foods. Pork liver contains more vitamin A, can nourish the eyeball, and has the effect of nourishing the liver and brightening the eyes. Suitable for myopia in children and adolescents.

51. [Chicken Liver Thick Soup] Food–medicine-homology ingredient: ginger.

Ingredients: 100 g chicken liver, appropriate amounts of salt, MSG, and ginger.

Method: Wash chicken liver, chop, and blanch in boiling water. When the chicken liver changes color and no blood remains, add minced ginger, salt, and MSG, and mix well.

Effects: Eat with meals. Chicken liver has a high vitamin A content. This recipe can nourish the liver and brighten the eyes, making it suitable for various types of myopia.

52. [Yogurt, Nut Butter, Carrot, Fruit, and Vegetable Salad] Food–medicine-homology ingredients: almond, honey, black pepper.

Ingredients: 1 carrot, appropriate amounts of other fruits and vegetables (such as apple, cucumber, cherry tomato, etc.), appropriate amount of yogurt, appropriate amount of nuts (such as almonds, walnuts, cashews, etc.), appropriate amount of honey, appropriate amounts of salt and black pepper (optional), appropriate amounts of fresh herbs (such as basil, parsley, etc., optional).

Method: (1) Wash and peel the carrot, cut into thin shreds or shave into thin slices with a peeler. Wash and cut other fruits and vegetables, keeping a uniform size for better mixing. Soak or roast nuts in advance (optional), then put them into a blender and crush. Add an appropriate amount of yogurt and honey, add a little salt and black pepper to season, and blend into a smooth nut butter. (2) Pour the blended yogurt nut butter over the cut carrot and other fruits and vegetables, and gently mix with chopsticks or a spoon.

Effects: Carrots are rich in β-carotene, which can be converted into vitamin A in the body and is crucial for maintaining vision health. Other vegetables and fruits in the salad also contain eye-beneficial nutrients such as vitamin C and lutein. Nuts in the yogurt and nut butter are rich in vitamin E, unsaturated fatty acids, etc., which helps eye health.

53. [Corn Beef Cheese Baked Five-Grain Rice] Food–medicine-homology ingredient: black pepper.

Ingredients: appropriate amount of corn kernels, beef, cheese, five-grain rice (mixed cooked rice of brown rice, white rice, oat rice, quinoa, millet, etc.), onion, carrot, green beans, tomato sauce, salt, black pepper, olive oil, etc.

Method: (1) Wash and mix brown rice, white rice, oat rice, quinoa, millet, etc. Add an appropriate amount of water and cook into five-grain rice, then let cool. Cut beef into small cubes, marinate with salt and black

pepper for 10 minutes; dice onion and carrot, wash green beans; cook or steam corn kernels in advance. (2) Heat a pan, add olive oil, add diced onion and stir-fry until fragrant; add beef cubes and stir-fry until the color changes; add diced carrots, green beans, and corn kernels and continue stir-frying; add an appropriate amount of tomato sauce and salt, stir-fry evenly, and turn off the heat. (3) Pour the stir-fried beef and vegetables into the cooled five-grain rice, gently mix with a spoon. (4) Put the mixed five-grain rice into a baking dish, spread it flat; sprinkle with an appropriate amount of shredded cheese; preheat the oven to 180°C, put the baking dish on the middle rack, and bake for 10–15 minutes until the cheese melts and turns golden brown.

Effects: Corn is rich in vitamin A, antioxidants (flavonoids, lutein, zeaxanthin), and vitamin C, helping maintain retinal function, protect eyes from free radical damage, and promote eye blood vessel health; beef provides high-quality protein and other nutrients; cheese supplies calcium and protein that support the structure of the eye

54. [Mushroom and Carrot Noodles]

Ingredients: 100 g noodles, 20 g each of mushroom and carrot, 100 g choy sum; appropriate amounts of garlic slices, salt, and vegetable oil.

Method: Wash choy sum and cut into sections; wash mushroom and carrot and slice. Heat oil in a pan, sauté garlic slices until fragrant, add carrot, mushroom, and choy sum, and stir-fry briefly. Add clean water and bring to a boil over high heat. Add noodles and cook until done. Add salt to season.

Effects: Mushrooms do not directly contain vitamin A, but under sunlight, they can convert to vitamin D, which has indirect benefits for eye health, such as promoting calcium absorption and helping maintain normal structure and function of eye tissues. Polysaccharides and other active ingredients in mushrooms also have antioxidant and anti-inflammatory effects, helping protect eyes from free radicals and other harmful substances. Carrots are rich in carotene, which can be converted into vitamin A in the body. Vitamin A is a key nutrient for maintaining normal visual function; it participates in the synthesis of rhodopsin in the retina, thereby preventing night blindness.

55. [Wheat Flour, Arborvitae Seed, and Poria Cake] Food–medicine-homology ingredient: poria.

Ingredients: 50 g coarse wheat flour, 15 g each of arborvitae seed and poria.

Method: Dry poria, stir-fry arborvitae seed until fragrant, grind together into powder, mix with wheat flour and water, and bake into cakes.

Effects: Suitable for myopia patients with heart and spleen deficiency type.

56. [Peanut, Melon Seed, Jujube, and Bean Cake] Food–medicine-homology ingredient: jujubes.

Ingredients: 100 g peanut kernels, 50 g pumpkin seeds, 60 g red date flesh, 30 g soybean flour, 150 g japonica rice flour.

Method: Put all ingredients into a small basin, add an appropriate amount of clean water and a little sesame oil, mix well to make a cake, and steam in a pot until cooked. Eat in portions throughout the day.

Effects: Tonifies the spleen and boosts qi, nourishes blood, and brightens the eyes. Mainly treats symptoms such as blurred vision, palpitations, shortness of breath, and constipation due to body deficiency.

57. [Walnut Kernels] Food–medicine-homology ingredients: black sesame, honey.

Ingredients: 1000 g walnuts, 500 g black sesame, 1 cup milk or soy milk, 1 tsp honey.

Method: Remove shells and skins from walnuts, stir-fry in an iron pan over low heat until slightly yellow, mash into a paste. Rinse black sesame with water, stir-fry in an iron pan over low heat until dry, and grind into

a fine powder. Take 1 tsp each of walnut paste and black sesame powder, mix into boiling milk or soy milk, let stand for a while, and after the temperature drops, add 1 tsp honey, mix well, and take once daily as a morning snack.

Effects: Long-term consumption can enhance ciliary muscle strength and scleral hardness, thereby preventing the occurrence and progression of myopia.

II. Visual Fatigue Category: Improving Blood Circulation, Accelerating Eye Metabolism, Combined with Appropriate Massage to Relieve Visual Fatigue

58. [Stir-Fried Lean Pork with Goji Leaves] Food–medicine-homology ingredient: goji leaves.

Ingredients: 250 g washed tender goji leaves, 250 g shredded lean pork.

Method: Stir-fry in oil, add salt, cooking wine, and other seasonings.

Effects: Eat with lunch and dinner. Suitable for myopia patients.

59. [Steamed Chicken with Astragalus and Ginkgo] Food–medicine-homology ingredients: astragalus, ginkgo, ginger.

Ingredients: 1 hen, 30 g astragalus, 6 g ginkgo, 10 g scallion, 5 g ginger, 3 g each of refined salt and pepper, 2 g MSG, 20 ml cooking wine, appropriate amount of fresh soup.

Method: (1) Slaughter chicken, remove feathers and intestines, wash, blanch in boiling water to remove blood, remove, and rinse with cold water. (2) Wash astragalus and ginkgo, soak in warm water for half an hour, and stuff into the chicken cavity. Place chicken on a plate, add an appropriate amount of fresh soup, add scallion sections, ginger slices, pepper, cooking wine, and refined salt, cover, and steam over high heat for 30 minutes. After removing from the steamer, pick out the astragalus, scallion sections, and ginger slices, and add MSG.

Effects: Boosts qi and consolidates the exterior, blackens hair, and brightens the eyes. Used for auxiliary treatment of premature graying of hair, hyperopia, myopia, and body deficiency prone to colds.

60. [Astragalus and Angelica Stewed Beef] Food–medicine-homology ingredients: astragalus, angelica, ginger, Sichuan pepper.

Ingredients: 200 g lean yellow beef, 50 g astragalus, 250 g tripe, 20 g angelica.

Method: Remove tendons from beef, cut into thin slices; wash tripe and slice. Decoct the two Chinese medicines in water and take the medicinal juice for later use. Place a wok over heat, add vegetable oil and heat to 60% hot, add minced ginger and Sichuan pepper and stir-fry a few times, add beef and tripe, then add fermented glutinous rice juice, refined salt, and chicken essence, bring to a boil, add fresh soup and bring to a boil, then add beef tallow and medicinal juice, bring to a boil, and simmer slowly.

Effects: Tonifies qi and blood, strengthens tendons and bones, and regulates the five viscera. Suitable for myopia patients with qi and blood deficiency.

61. [Astragalus Stewed Hen] Food–medicine-homology ingredient: astragalus.

Ingredients: 500 g hen meat, 60 g honey-fried astragalus slices (wrapped in cloth).

Method: Stew together, remove astragalus.

Effects: Take in appropriate amounts. Suitable for myopia patients with qi and blood deficiency type.

62. [Steamed Goji Berries with Longan] Food–medicine-homology ingredients: goji berries, longan.

Ingredients: 7 goji berries, 1 longan.

Method: Embed 7 goji berries into one longan, steam 7 pieces daily, and take.

Effects: Can prevent or reduce the occurrence of myopia.

63. [Longan and Codonopsis Porridge] Food–medicine-homology ingredients: longan flesh, codonopsis.

Ingredients: 150 g japonica rice, 30 g codonopsis, 15 g longan flesh.

Method: Decoct codonopsis in water, take the juice, and cook porridge with longan flesh and japonica rice.

Effects: Divide into 2 servings. Suitable for myopia patients with qi and blood deficiency type.

64. [Sweet Potato Porridge]

Ingredients: 250 g fresh sweet potato, 100–150 g japonica rice, appropriate amount of white sugar.

Method: Wash sweet potato, cut into small pieces with skin, wash japonica rice with water; place a pot on heat, add water, sweet potato pieces, and japonica rice, cook together into porridge. When the porridge is almost done, add an appropriate amount of white sugar, bring to a boil 2–3 times, and serve.

Effects: Sweet potato tonifies deficiency, boosts strength, strengthens the spleen and stomach, and strengthens kidney yin. Sweet potato does not directly act on the eyes to relieve visual fatigue, but solves the problem fundamentally by enhancing organ function.

65. [Black Chicken Porridge] Food–medicine-homology ingredient: ginger.

Ingredients: 100 g black chicken meat, 200 g japonica rice. Appropriate amounts of scallion, ginger, refined salt, pepper, and sesame oil.

Method: Put black chicken meat and japonica rice into a pot, season with scallion, ginger, refined salt, pepper, and sesame oil, and cook into porridge.

Effects: Nourishes the liver and kidneys, benefits essence, and nourishes blood. Suitable for myopia of liver and kidney deficiency type, with symptoms of difficulty seeing near objects, dizziness, frequent dreams at night, and soreness and weakness of the waist and knees.

66. [Goji Leaf Egg Drop Soup]

Ingredients: 200 g goji leaves, 2 eggs, appropriate amounts of oil, salt, and chicken essence.

Method: Wash goji leaves, beat eggs. Heat oil in a pan, first stir-fry eggs, then add two bowls of clean water and bring to a boil; add goji leaves, add appropriate amounts of salt and chicken essence to season, and serve.

Effects: Goji leaves are actually the tender leaves of the well-known Chinese herb goji. They also have the effects of tonifying deficiency, benefiting essence, clearing heat, and brightening the eyes, and can be used to treat dry eyes and fatigue. Modern research has also found that goji leaves are not only rich in various trace elements but also rich in vitamins and more than a dozen essential amino acids. Regular consumption can not only protect the eyes but also prevent and treat constipation, cough, dry throat, thirst, and other sub-health problems.

67. [Goji Berry, Chenpi, and Longan Drink] Food–medicine-homology ingredients: goji berries, chenpi, longan, honey.

Ingredients: 10 g goji berries, 3 g chenpi, 10 longan flesh, 1 tsp honey.

Method: Put goji berries and chenpi in a bag made of two layers of gauze, then put them with longan flesh in a pot, add an appropriate amount of water, simmer over low heat for half an hour, take out the longan flesh and soup, and add honey.

Effects: Goji berries are also rich in carotene, which can relieve dizziness, blurred vision, and other symptoms caused by liver and kidney yin deficiency. Chenpi has the effects of dispersing lung qi and regulating the middle, and directing counterflow downward. Combined with goji berries, it helps relieve cough, excessive phlegm, and asthma caused by lung dryness. Longan flesh contains a large amount of glucose, fructose, fruit acids, multiple vitamins, trace elements, and minerals, and can tonify qi and nourish blood, nourish yin and tonify the kidneys, soothe the liver and relieve depression, and strengthen the stomach and spleen. This drink mainly improves vision by nourishing the liver and kidneys and brightening the eyes.

68. [Three-Treasure Drink] Food–medicine-homology ingredients: goji berries, chenpi, longan flesh, honey.

Ingredients: 20 g goji berries, 5 g chenpi, 8 longan flesh, 1 tbsp honey.

Method: Put goji berries, chenpi, and longan flesh in a pot, add an appropriate amount of water, bring to a boil over high heat, then simmer over low heat for 20 minutes. After cooling, add honey and take.

Effects: Once daily in the afternoon. Suitable for adolescent myopia patients.

69. [Blueberry Milk Cake]

Ingredients: Main materials: 5 tbsp dried blueberries, 220 ml milk, 35 g corn starch, an appropriate amount of coconut shreds, a little crushed rose petals, half a bowl of clean water.

Method: Add half a bowl of water to cornstarch, mix until uniform and free of lumps, and set aside. Pour 220 ml of milk into a blender, add dried blueberries, and soak for 15 minutes. After soaking blueberries, blend until smooth, then filter. Take a pot, pour in the milk-blueberry liquid, heat over low heat, and slowly add cornstarch slurry. Stir continuously, cook over low heat until cooked through, thick, and smooth. Pour the milk-blueberry paste into a mold, spread it flat as much as possible, and let it cool at room temperature until solidified. Turn out the blueberry milk cake, sprinkle with crushed rose petals and coconut shreds, cut into small squares, and serve.

Effects: Blueberries are rich in anthocyanins, a powerful antioxidant. Anthocyanins can activate retinal function and promote the regeneration of rhodopsin in retinal cells. Rhodopsin is an important substance in the eye, crucial for night vision and adaptation to low-light environments. By promoting rhodopsin regeneration, blueberries help improve vision and prevent myopia. Anthocyanins can also improve ocular microvascular blood supply and increase ocular blood circulation.

III. Inflammatory Injury Category: For Eye Inflammation Caused by Excessive Eye Use, Visual Fatigue, etc.,

Chrysanthemum and similar herbs with heat-clearing, detoxifying, and swelling-reducing effects can be used, but long-term or high-dose use is not recommended as it may affect constitution

70. [Cold Dressed Shepherd's Purse]

Ingredients: 500 g shepherd's purse, 100 g dried tofu, appropriate amounts of sesame oil, soy sauce, and vinegar.

Method: Wash shepherd's purse, blanch, chop; dice dried tofu, mix with chopped shepherd's purse, drizzle with sesame oil, soy sauce, and vinegar.

Effects: Shepherd's purse has the effects of harmonizing the spleen, promoting diuresis, brightening the eyes, and stopping bleeding. It can play an auxiliary conditioning role in edema, dysentery, red and swollen eyes, and other conditions. Modern research has found that shepherd's purse is rich in β-carotene, vitamin C, and B

vitamins. β-carotene can be converted into vitamin A in the body, which not only benefits eye and skin health but is also an indispensable nutrient for many defense systems against infection in the body.

71. [Stir-Fried Pork Slices with Chrysanthemum] Food–medicine-homology ingredients: chrysanthemum, ginger.

Ingredients: 500 g lean pork, 100 g fresh chrysanthemum petals, 3 eggs, a little chopped scallion and minced ginger.

Seasonings: appropriate amounts of salt, MSG, cooking wine, vegetable oil, and starch.

Method: (1) Wash pork and slice; gently wash chrysanthemum petals; crack eggs into a bowl, add cooking wine, salt, and starch to make a batter, add pork slices and mix well. (2) Heat vegetable oil in a pan, add pork slices and cook until cooked through, remove. (3) Leave a little oil in the pan, add chopped scallion and minced ginger and sauté until fragrant, add pork slices, water, and chrysanthemum petals, stir-fry evenly, season with MSG, mix well, and serve.

Effects: Chrysanthemum can dispel wind and clear heat, nourish the liver, and brighten the eyes. Lean pork is rich in vitamin B1 and zinc, and eggs contain vitamin A and vitamin B1. This dish has the effects of dispelling wind, clearing heat, calming the liver, and brightening the eyes, beneficial to eye health.

72. [Stir-Fried Pork Heart with Goji Leaves]

Ingredients: 50 g fresh goji leaves, 1 pig heart, an appropriate amount of peanut oil, a little salt.

Method: Heat peanut oil, add sliced pig heart and goji leaves, stir-fry until cooked, add salt to season, and serve.

Effects: Tonifies the liver and benefits essence, nourishes the heart and calms the mind, clears heat, and brightens the eyes. Used for myopia of yin deficiency with internal heat type.

73. [Stewed Hen with Mulberry and White Chrysanthemum] Food–medicine-homology ingredients: mulberry, chrysanthemum.

Ingredients: 50 g mulberry, 30 g white chrysanthemum, 1 hen, appropriate amounts of cooking wine, salt, soy sauce, and MSG.

Method: Slaughter and wash the hen, put mulberry and chrysanthemum into the chicken cavity; put the chicken into a clay pot, add cooking wine, salt, soy sauce, MSG, and clean water, bring to a boil over high heat, switch to low heat and simmer until the chicken is cooked, remove the medicinal residue, and eat.

Effects: Suitable for myopia patients.

74. [Stir-Fried Daylily with Wood Ear]

Ingredients: 20 g black fungus, 80 g daylily, a little chopped scallion.

Seasonings: appropriate amounts of salt, MSG, vegetable oil, and water starch.

Method: (1) Soak black fungus in warm water, remove impurities, wash, and tear into small pieces; soak daylily in cold water, remove impurities, wash, squeeze out water, and cut into small sections. (2) Heat vegetable oil in a pan, add chopped scallion and sauté until fragrant, add daylily and black fungus and stir-fry, add water, salt, and MSG, stir-fry until vegetables are cooked and flavorful, thicken with water starch, and serve.

Effects: Black fungus polysaccharides have anti-aging and immunity-enhancing effects, and can also scavenge free radicals and protect eyes from oxidative damage. Daylily contains a large amount of carotene, which can be converted into vitamin A in the body, helping maintain clear vision and prevent night blindness and dry eye disease.

Contraindications: Daylily should be a dried product. Fresh daylily contains colchicine, which, after being fried and eaten, can be oxidized in the body to produce a highly toxic substance. Generally, poisoning symptoms appear 30 minutes to 7 hours after eating, with mild cases experiencing nausea and vomiting, and severe cases experiencing abdominal pain, diarrhea, and in serious cases, blood in urine and stool.

75. [Egg-Sliding Bitter Melon Slices]

Ingredients: 1 bitter melon, 2 eggs.

Seasonings: appropriate amounts of salt, MSG, and vegetable oil.

Method: Remove seeds from bitter melon, wash, and cut into thin slices; crack eggs into a bowl, add salt, and beat evenly. Heat vegetable oil in a pan, add bitter melon slices, and stir-fry until cooked. Add more oil to the pan, heat, pour in egg liquid and bitter melon slices, push evenly with a spoon, stir-fry until eggs are cooked, add MSG and salt, and serve.

Effects: Bitter melon is a nutrient-rich vegetable, containing quinine, protein, fat, carbohydrates, and rich vitamins C, B1, B2, calcium, phosphorus, iron, and other nutrients. In traditional Chinese medicine, bitter melon is believed to have the effects of "removing pathogenic heat, relieving fatigue, clearing the heart, and brightening the eyes." *Compendium of Materia Medica* also records bitter melon's heart-clearing and eye-brightening effects. In addition, vitamin C and B vitamins in bitter melon are particularly important for eye health; they help maintain retinal health and promote normal operation of the optic nerve. Bitter melon paired with eggs is beneficial for protecting the eyes and can reduce the occurrence of eye diseases.

76. [Boiled Eggs with Cassia Seed] Food–medicine-homology ingredient: cassia seed.

Ingredients: 60 g cassia seed, 5 eggs.

Method: Boil eggs, remove shells, and cook with cassia seed.

Effects: Eat. Suitable for myopia patients.

77. [Goji Berry, Chrysanthemum, and Lean Pork Porridge] Food–medicine-homology ingredients: goji berries, chrysanthemum, ginger.

Ingredients: 5 g each of goji berries and chrysanthemum; 50 g lean pork, 100 g rice, appropriate amounts of scallion, ginger, Sichuan pepper, refined salt, and MSG.

Method: Wash lean pork, cut into fine pieces, put into a pot with goji berries and rice, add an appropriate amount of clean water, and cook into thin porridge. When nearly done, add chrysanthemum, chopped scallion, minced ginger, Sichuan pepper, refined salt, and MSG, bring to a boil once or twice, and serve.

Effects: Nourishes the liver and kidneys, nourishes yin, and brightens the eyes.

78. [Red Date and Chrysanthemum Porridge] Food–medicine-homology ingredients: jujubes, chrysanthemum, honey.

Ingredients: 100 g rice, 20 g red dates, 10 g chrysanthemum, 1 tbsp honey (about 15 ml).

Method: (1) Wash rice (after washing, soak in clean water for 30 minutes). Wash red dates and remove pits, wash chrysanthemum in warm water and soak for later use. (2) Pour an appropriate amount of clean water into a pot, put rice and red dates together, after water boils, switch to low heat and cook for 15 minutes, then add chrysanthemum and cook together for 10 minutes. After removing from heat and cooling slightly, mix in honey and eat.

Effects: Red dates have always been a good product for tonifying qi and nourishing blood; chrysanthemum has the effects of dispersing wind, clearing heat, brightening the eyes, and detoxifying. The two combined in porridge can make skin beautiful, eyes bright, and complexion excellent.

79. [Purslane and Pork Liver Porridge] Food–medicine-homology ingredient: purslane.

Ingredients: 45 g purslane, 30 g each of enoki mushroom and pork liver, 3 eggs, appropriate amount of salt.

Method: First wash purslane, cut into small sections; tear enoki mushrooms into strips; wash pork liver and cut into thin slices. Add water to a pot, put in purslane and enoki mushrooms, and cook for 15 minutes; add pork liver and simmer briefly, crack in eggs, and after the water boils, add an appropriate amount of salt to season.

Effects: Pork liver itself has the effect of nourishing the liver and brightening the eyes. Adding purslane and enoki mushrooms makes a soup that is not only tender and fragrant but also has the effects of benefiting the liver, brightening the eyes, and soothing the middle and descending qi. Purslane is a food–medicine-homology wild herb, cold in nature, entering the liver meridian, and can cool blood and reduce liver fire. This medicinal diet is suitable for eye diseases caused by liver fire flaming upward.

80. [Shepherd's Purse Porridge]

Ingredients: 100 g shepherd's purse, 100 g japonica rice.

Method: First wash shepherd's purse, cut into small pieces and set aside; put japonica rice in a pot, add an appropriate amount of clean water, and cook porridge. After the rice is cooked and soft, add shepherd's purse pieces and cook for another 2 minutes.

Effects: Has the effects of clearing heat, brightening the eyes, stopping bleeding, and strengthening the spleen, helping adolescents prevent and treat myopia.

81. [Kelp and Cassia Seed Soup] Food–medicine-homology ingredient: kelp.

Ingredients: 25 g kelp, 12 g cassia seed.

Method: Soak kelp in water until soft, wash, and cut into shreds. Put into a pot, add an appropriate amount of water, and cook into soup with cassia seed.

Effects: Eat on an empty stomach in the morning, eat kelp and drink soup. Clears the liver and brightens the eyes.

82. [Goji Berry and Chrysanthemum Tea] Food–medicine-homology ingredients: goji berries, chrysanthemum.

Ingredients: goji berries, chrysanthemum.

Method: Brew with hot water and drink. Can be re-brewed, but do not re-brew overnight. You can also cook porridge with goji berries, which has obvious effects on improving blurred vision and excessive tearing.

Effects: Goji berries are rich in carotene (provitamin A), vitamins B1, B2, and calcium, which are essential nutrients for healthy eyes. Long-term consumption can make eyes relaxed and bright.

83. [Chrysanthemum Tea] Food–medicine-homology ingredient: chrysanthemum.

Ingredients: Choose chrysanthemums with small flowers and a yellowish color.

Method: Brew in water or boil and drink. Drink it hot in winter and iced in summer.

Effects: Drinking three or four cups of chrysanthemum tea daily can not only eliminate eye fatigue symptoms but also help restore vision.

Appendix 5: 55 Questions on Dietary Therapy Knowledge for Myopia Prevention

I. Foods to Avoid for Myopia Prevention

[1] Why is garlic bad for the eyes?

Answer: Garlic damages the liver and eyes and can aggravate eye diseases. Moreover, pungent foods like garlic may damage the liver and kidneys, so they should be eaten sparingly.

[2] What are spicy and stimulating foods, and what harm does eating too much do to the eyes?

Answer: Chili, curry, mustard, garlic, and scallion are spicy and stimulating foods. If children eat too much of these, the body easily develops internal heat, the eyes may have a burning sensation, and blood vessels in the eyeball may become congested, and it may cause conjunctivitis and decreased vision.

[3] What is the relationship between overcooked protein foods and vision?

Answer: Eating overcooked protein foods can cause abnormal calcium metabolism in the body, leading to calcium deficiency. Calcium is very important for normal eye development and function; calcium deficiency may indirectly affect vision.

[4] What adverse effects does sugar have on vision?

Answer: Sugar metabolism in the body requires a large amount of vitamin B. Excessive sugar intake leads to vitamin B deficiency, affecting optic nerve growth and development and causing decreased vision. In addition, eating too much sugar reduces calcium in the body, weakens eyeball elasticity, reduces eye tissue elasticity, decreases storage of trace element chromium, and easily lengthens the axial length. When axial length exceeds the normal range, especially greater than 24 mm, it is generally considered myopia clinically. Elevated blood sugar causes changes in aqueous humor and lens osmotic pressure, prompting the lens anteroposterior diameter to lengthen, causing myopia. Aqueous humor is a transparent clear fluid filling the anterior and posterior chambers of the eye. The main function of aqueous humor is to maintain intraocular pressure, maintain the shape of the eyeball, provide necessary nutrition to the cornea and lens, and help excrete metabolic products from the eye. Lens osmotic pressure mainly refers to the osmotic pressure produced by low-molecular-weight crystalline substances in plasma (such as sodium chloride, sodium bicarbonate, glucose, inorganic salts, etc.). Normal lens osmotic pressure is crucial for maintaining normal cell morphology and function. For example, when blood sugar rises, it causes changes in aqueous humor and lens osmotic pressure, which may affect the shape of the lens, causing its anteroposterior diameter to lengthen and leading to myopia. Increased blood sugar can also change blood acidity, affect calcium absorption, lead to body calcium deficiency, and adversely affect vision.

[5] Why are desserts not conducive to myopia prevention?

Answer: Desserts cause calcium loss and consume B vitamins in the body. Because sugar in sweets requires a large amount of vitamin B for metabolism, if children eat too much sugar, their vitamin B is insufficient. Moreover, excessive sugar intake reduces calcium in the body, weakens eyeball elasticity, and is not conducive to normal vision development; instead, it accelerates myopia formation.

[6] What adverse effects does fat have on the eyes?

Answer: Fat is called “fatty paste” in traditional Chinese medicine. Eating too much generates dampness and transforms into fire. Limiting intake of lard and high-protein animal fat can reduce chromium excretion from the body, which is beneficial for preventing and treating myopia. Therefore, fried foods should not be eaten too much.

[7] What adverse effects do foods high in oxalic acid have?

Answer: Houttuynia, amaranth, spinach, and pokeberry contain more oxalic acid. Oxalic acid affects the body's absorption of calcium, and calcium plays a key role in the development and function of eye-related tissues such as the sclera. Insufficient calcium absorption may affect vision.

[8] What adverse effects do acidic foods have?

Answer: Do not eat too many acidic foods. If you eat too much acidic food or sugar, an acidic environment forms in the gastrointestinal tract, causing calcium in food to become calcium salts that are excreted in feces, and the body cannot absorb enough calcium. The development of the human eye sclera requires sufficient calcium. If calcium is insufficient, the sclera is not firm, and its hardness and toughness are weak, making it prone to causing or aggravating myopia. This is a major issue in myopia prevention that must be taken seriously.

II. Foods Recommended for Myopia Prevention

[9] What benefits do goji berries have for the eyes?

Answer: Goji berries can clear the liver and brighten the eyes, help improve pseudomyopia, and inhibit myopia progression.

[10] Why are animal livers beneficial for eye health?

Answer: Animal livers, such as pork liver, chicken liver, and sheep liver, are rich in vitamin A. This nutrient can protect the eyes, maintain normal vision, and has the effect of tonifying the liver, nourishing the liver, and brightening the eyes. They are good ingredients for protecting the eyes.

[11] What benefits do fish have for the eyes?

Answer: Fish are rich in amino acids and various trace elements. These nutrients can provide sufficient nourishment for the eyes and keep them healthy.

[12] What is unique about sea fish in protecting vision?

Answer: Australian researchers found that eating fish can prevent age-related macular degeneration, which can lead to blindness and is difficult to cure. Fish is rich in zinc and protein. Eating an appropriate amount of sea fish once a week can protect the eyes.

[13] What is the relationship between milk and vision?

Answer: Milk is rich in calcium. Calcium can improve eye fatigue and protect vision, so drinking more milk is good for the eyes.

[14] Why are carrots good for the eyes?

Answer: Carrots benefit the liver and brighten the eyes. They are rich in vitamin A, which can protect the eyes and allow them to function normally.

[15] What similarities do red radish and cod liver oil have in eye protection?

Answer: Red radish contains provitamin A, which can be converted into vitamin A in the body. So its effect is similar to cod liver oil—both provide beneficial nutritional support for the eyes and help prevent some eye problems such as corneal dryness and decreased vision.

[16] Does eating hard foods have special effects on the eyes?

Answer: Yes! Hard foods such as carrots, fruits, and nuts can exercise the eye muscles well during chewing, like doing eye exercises, and can improve the eye's self-regulation ability. Japanese researchers found that people who often eat soft foods are more likely to have decreased vision than those who prefer hard foods, so chewing is called "another kind of eye exercise."

[17] What effect do strawberries have on vision?

Answer: Strawberries can both protect vision and help maintain the liver, indirectly benefiting eye health.

III. Food Types and Myopia Prevention

[18] What foods brighten the eyes and tonify the liver?

Answer: Traditional Chinese medicine says myopia is related to liver and kidney qi and blood deficiency, so eating foods that brighten the eyes and tonify the liver can help prevent and treat myopia. For example, eating fish or poultry eye tissues is good because they are rich in various proteins, amino acids, and trace elements, especially selenium. Stewing in clear water preserves nutrients, and pairing it with black fungus is even better. In addition, the Compendium *of Materia Medica* mentions that the eyes of a mynah or eagle ground with milk and dripped into the eyes can brighten vision, but this method is not commonly used.

[19] Why can myopia prevention and treatment choose foods that tonify the liver and kidneys?

Answer: Traditional Chinese medicine believes myopia is mainly caused by liver and kidney insufficiency and qi and blood deficiency. Foods such as meat, eggs, liver, kidney, crucian carp, yellow croaker, cuttlefish, sea cucumber, soft-shelled turtle, and longan, lychee, grape, walnut, mulberry, and jujube all have the effect of tonifying the liver and kidneys, helping improve myopia caused by liver and kidney insufficiency.

[20] What foods strengthen the spleen and stomach and tonify qi and blood?

Answer: Carrot, sweet potato, longan flesh, yam, taro, spinach, millet, corn, and other foods can strengthen the spleen and stomach and tonify qi and blood, and also support eye health to some extent, because sufficient qi and blood are important for nourishing the eyes.

[21] Why eat more alkaline foods to prevent myopia?

Answer: If acidic foods are consumed in excess, it easily causes myopia. To prevent and treat myopia, acid-base balance should be maintained, so more alkaline foods should be eaten. Alkaline foods are mainly vegetables, fruits, and beans, such as brown rice, beans, green and yellow vegetables like rapeseed, carrot, green pepper, peas, radish greens, light-colored vegetables like radish, cabbage, kale, cucumber, algae like nori and wakame, and sesame.

IV. Antioxidant Vitamins and Myopia Prevention

[22] What effects do antioxidant vitamins have on the eyes?

Answer: Antioxidant vitamins mainly include vitamins A, C, and E. These vitamins play important antioxidant roles in the body, helping scavenge free radicals and protect cells from oxidative damage.

[23] What important role does vitamin A play for the eyes?

Answer: Vitamin A can prevent corneal dryness and degeneration, eliminate eye fatigue, and has a significant effect on preventing vision decline in children. It can also synthesize rhodopsin in the eye, which is very important for maintaining normal good vision adaptation and is also related to maintaining epithelial cell

structural integrity and promoting growth and development. Vitamin A deficiency can cause night blindness, dry eye disease, and corneal softening, manifested as unclear vision in dim light, dry eyes, and easy fatigue.

[24] Why is vitamin A called the "eye protection guard" for computer users?

Answer: Many office workers today spend too long in front of monitors, and rhodopsin in the retina is consumed in large amounts. Rhodopsin is mainly synthesized from vitamin A. So, vitamin A can prevent corneal dryness and degeneration, eliminate eye fatigue, and prevent vision decline, which is crucial for computer users to protect their eyes. We can supplement vitamin A by eating more carrots, red dates, oranges, tomatoes, and other fruits and vegetables, or by eating deep-sea fish and cod liver oil.

[25] Which foods are rich in vitamin A?

Answer: Animal livers, cod liver oil, fish roe, whole milk and whole milk products, and eggs have high vitamin A content and are good eye-protecting foods. Sheep liver has long been used as a folk remedy for eye diseases. In addition, plant foods such as carrots, red-hearted sweet potatoes, apricots, persimmons, and tomatoes contain provitamin A (carotene), which can be converted into vitamin A in the body. Some red and orange fruits and vegetables, such as carrots, red peppers, crabapples, orange and yellow tomatoes, oranges, lemons, and grapefruits, not only contain vitamin A but are also rich in vitamin C needed for vision. Egg yolks, milk, lean meat, spinach, Chinese cabbage, and fish also contain vitamin A.

[26] What effects does vitamin E have on the eyes?

Answer: Vitamin E has antioxidant effects, can inhibit lipid peroxidation reactions in the lens, dilate peripheral blood vessels, improve blood circulation, promote recovery of diseased tissues, and benefit eye health.

[27] Which foods are high in vitamin E?

Answer: Vitamin E is high in plant seeds such as peanuts, walnuts, and pine nuts. In the diet, soybean oil, peanut oil, and bananas also have considerable vitamin E content. Every 100 g of peanut oil contains 12 mg of vitamin E, one of the main dietary sources of vitamin E.

[28] What benefits does vitamin C have for the eyes?

Answer: Vitamin C can prevent eyes from being damaged by sunlight's ultraviolet rays and can also prevent free radical damage to eye tissues. The U.S. National Institutes of Health (NIH) has also scientifically evaluated the eye health benefits of taking antioxidants.

[29] Which foods are rich in vitamin C?

Answer: Dark green vegetables and various fruits are rich in vitamin C.

Among fruits, wild jujube has the highest vitamin C content, 1170 mg per 100 g; sweet jujube is next; and grapefruit, hawthorn, longan, persimmon, orange, lemon, and strawberry also contain it. Among vegetables, chili has the highest vitamin C content, up to 198 mg per 100 g; bell pepper is next; and kale, garlic sprouts, cauliflower, snow mustard, celery, tatsoi, tomato, chives, and Chinese artichoke are also rich in it.

V. Vitamins B and D and Myopia Prevention

[30] What is the importance of vitamin B1 for the eyes?

Answer: Vitamin B1 is one of the nutritional sources of the optic nerve. It plays a key role in nerve conduction because acetylcholine is important in nerve conduction. Vitamin B1 deficiency hinders acetylcholine production, causing optic nerve conduction disorders, affecting normal function of the cornea and refractive adjustment muscles, leading to decreased vision. When vitamin B1 is deficient, the eyes may experience dryness,

decreased vision, dilated pupils, slow light response, pulling pain during eye movement, and deep orbital pressure and pain. This is because it mainly promotes cell metabolism, especially closely related to nerve conduction function.

[31] Which foods contain more vitamin B1?

Answer: Coarsely processed rice and flour, miscellaneous grains, soy products, animal organs, and lean meat contain more vitamin B1.

[32] What effects does vitamin B2 deficiency have on the eyes?

Answer: Vitamin B2 deficiency often causes optic neuritis, blepharitis, conjunctivitis, and other eye diseases, manifested as decreased vision, photophobia, tearing, and conjunctival congestion.

[33] Which foods are rich in vitamin B2?

Answer: Yeast, dairy products, beef, animal liver, soybeans, spinach, amaranth, wood ear, and sunflower seeds are rich in vitamin B2.

[34] What is the relationship between vitamin D and myopia prevention?

Answer: Vitamin D has an indirect effect on myopia prevention because calcium absorption and utilization require vitamin D, and calcium and chromium both affect the regulation of fluid pressure in the eye, thereby relating to myopia prevention.

[35] Where does vitamin D come from?

Answer: The main source of vitamin D is not food but 7-dehydrocholesterol in the skin, which is converted by ultraviolet rays in sunlight. Therefore, to prevent calcium deficiency, we need to often be outdoors and receive sunlight.

VI. Provitamin and Myopia Prevention

[36] What is provitamin A?

Answer: Provitamin refers to substances that are not vitamins themselves but can be converted into vitamins in the body. Provitamin A usually refers to carotene, such as β-carotene, a common provitamin A. It can be converted into vitamin A in the human body, thereby exerting vitamin A's physiological functions, which are important for maintaining vision and promoting growth and development.

[37] What benefits does provitamin A have for the eyes?

Answer: After provitamin A is converted into vitamin A, it can prevent corneal dryness and degeneration, eliminate eye fatigue, and prevent vision decline. Lutein and β-carotene in various green vegetables (especially broccoli and leafy greens) can also prevent eyes from ultraviolet damage and free radical damage to eye tissues. However, the absorption and utilization rate of provitamin A is relatively low; the effect of 1 mg carotene is approximately equivalent to 278–556 international units of vitamin A.

[38] How is provitamin A converted into vitamin A in the body?

Answer: Provitamin A must undergo chemical decomposition in the human body before it can be converted into vitamin A, just like a seed germinating and growing in a suitable environment and finally becoming fruit. Through a series of changes, provitamin A can exert vitamin A's effects.

[39] Which foods are rich in provitamin A?

Answer: Many foods are rich in provitamin A, especially green vegetables, yellow vegetables, and fruits, such as celery, spinach, chives, pea shoots, alfalfa, green pepper, sweet potato, tomato, carrot, pumpkin, apricot,

and mango. Among them, the yellower the flesh of the sweet potato, the higher the carotene content, up to 5.51 mg per 100 g. Various green leafy vegetables such as perilla leaves, winter amaranth, celery leaves, mustard, and spinach are also rich in it, with darker colors indicating higher content. Various wild vegetables, such as qianbuiqi, have carotene content as high as 12.6 mg per 100 g, an excellent source of vitamin A. In addition, eating fish roe, fish meat, and fish liver (cod liver is best) also provides vitamin A and carotene that promote vision recovery.

VII. Protein and Myopia Prevention

[40] Which foods are rich in protein?

Answer: Lean pork, beef, mutton, chicken, and duck, animal organs, milk, fish, and beans are rich in protein. In addition, fish, meat, milk, and eggs are also rich in protein.

[41] What effect does excessive protein intake have on myopia prevention?

Answer: Eating excessive protein causes loss of vitamin B in the body, leading to calcium deficiency. Calcium, vitamin B, and other nutrients are important for normal eye development and function. Calcium deficiency or vitamin B insufficiency may indirectly affect myopia prevention, so protein intake should be moderate.

VIII. Lutein and Myopia Prevention

[42] What important role does lutein play for the eyes?

Answer: Lutein is an oxygen-containing carotenoid. Lutein has an important protective effect on the macula of the retina. The macula is the key part for seeing; like the film in a camera, if the film is damaged, it cannot sense light and cannot form clear images. Once macular degeneration occurs, vision decreases and may even be lost. Lutein can also prevent eyes from ultraviolet damage and free radical damage to eye tissues. If children lack lutein, it can lead to decreased vision.

[43] Where is lutein generally found?

Answer: Lutein is a carotenoid, abundant in fresh green leafy vegetables and citrus fruits, such as marigold, cauliflower, pumpkin, spinach, and parsley. In particular, green vegetables and broccoli contain more lutein and β-carotene.

IX. Zinc and Myopia Prevention

[44] What important role does zinc play for the eyes?

Answer: Zinc and protein are indispensable for retinal health. Zinc is abundant in the eye; it combines with melanin in the iris, improving the eye's adaptability to dim light and protecting vision. Zinc also participates in vitamin A metabolism and plays an important role in maintaining the tissue function of retinal pigment epithelial cells. Zinc deficiency may cause dull eyes and visual impairment. Some experts believe age-related macular degeneration is also related to zinc deficiency.

[45] Which foods are rich in zinc?

Answer: Many foods are rich in zinc. For example, coarse grains, coarse flour, millet, sweet potato, soybeans, milk, goat milk, beef, mutton, pork, fish, etc. Also, animal liver and kidneys, seafood, dairy, grains, beans, nuts—such as soybeans, oat flour, almonds, nori, kelp, mutton, steak, yellow croaker, oysters, milk powder, cocoa

powder, and tea—contain more zinc. In addition, seafood, whole grains, garlic, onion, longan, and goji berries also contain certain amounts of zinc. Oysters and apples are relatively rich in zinc.

X. Selenium and Myopia Prevention

[46] What benefits does selenium have for the eyes?

Answer: Selenium can enhance the eyes' resistance to bacteria and viruses, helping prevent eye diseases. Eating fish or poultry eye tissues has a good effect on preventing and treating myopia because they are rich in selenium and other trace elements, providing good nutritional support for the eyes. When cooking animal eye tissues, clear stewing can maintain higher nutritional content; adding black fungus makes the effect even better.

[47] Which foods are rich in selenium?

Answer: Many foods are rich in selenium. Such as fish, shrimp, and shellfish aquatic products, as well as noodles, brown rice, soybeans, sesame, chili, garlic, onion, mushroom, shepherd's purse, carrot, Chinese cabbage, etc. In addition, bread made from flour with bran also contains a large amount of selenium. Animal eye tissues are also a good source of selenium; besides containing various proteins and amino acids, their selenium content is higher than other tissues.

XI. Chromium and Myopia Prevention

[48] What is the relationship between chromium and myopia prevention?

Answer: Chromium is one of the essential trace elements for the human body. It plays an important role in maintaining normal metabolism and physiological functions. Chromium can combine with globulin to form biologically active complexes and participate in the metabolism of sugars, fats, and proteins. Many scientists have found that the occurrence of myopia (especially high myopia) is closely related to poor dietary habits and deficiency of calcium and chromium. Medical experts believe calcium and chromium can directly affect the body's regulation of fluid pressure in the eye. If fluid pressure in the eye remains normal, it helps prevent myopia. Examinations have also found that myopia patients generally lack chromium and calcium in their hair, so chromium and calcium deficiency are closely linked to the formation of myopia and the deepening of myopia degree.

[49] Which foods are rich in chromium?

Answer: Many foods contain chromium. Such as rice, corn, wheat, and radish and other vegetables; also brown rice, coarse flour, millet, corn, animal liver, lean meat, fish and shrimp, eggs, shellfish, and fungi; green leafy vegetables and fresh fruits also contain chromium. Yeast, beef, grains, meat, liver, and cheese contain more chromium. In addition, egg yolk, whole grains, pork, nuts, mushrooms, mutton, lean pork, chicken, beef, milk powder, and beans are also rich in chromium. In short, lean meat, eggs, fruits, and vegetables are all chromium-rich foods.

XII. Calcium and Myopia Prevention

[50] What role does calcium play in myopia prevention?

Answer: Calcium and phosphorus are important trace elements for myopia prevention. Scientists have found that the occurrence of myopia (especially high myopia) is closely related to poor dietary habits and deficiency of calcium and chromium. Calcium and chromium can directly affect the body's regulation of fluid

pressure in the eye. Maintaining normal fluid pressure in the eye is an important factor in preventing myopia. Examinations of myopia patients show that they generally lack chromium and calcium in their hair, indicating that chromium and calcium deficiency is closely linked to the formation of myopia and the deepening of myopia degree. Therefore, ensuring adequate calcium intake is of great significance for myopia prevention.

[51] Which common foods are rich in calcium?

Answer: Many foods are rich in calcium. For example, milk and dairy products have high calcium content and are easily absorbed and utilized by the body. Ordinary milk can contain up to 122 mg calcium per 100 g, and high-calcium milk has even more. Beans and their products, such as soybeans, black beans, edamame, traditional tofu, and dried tofu, are also good sources of calcium; dried tofu can contain up to 986 mg of calcium per 100 g. Seafood, such as shrimp, crab, clam, oyster, yellow croaker, dried small shrimp, jellyfish, kelp, nori, moss,

Seaweed and other seafood are rich in calcium. Sesame has the highest calcium content among plant foods, up to 564 mg per 100 g. Also, egg yolk, cheese, sesame paste, and vegetables such as rapeseed, amaranth, water spinach, celery, shepherd's purse, Chinese toon, Indian aster, wood ear, thistle, and water celery; animal bones such as beef bones and pig bones are also rich in calcium and easily absorbed. In addition, fish, dried shrimp, walnuts, and peanuts also contain considerable calcium.

XIII. Cooking Methods and Myopia Prevention

[52] What are the requirements for cooking vegetables?

Answer: When cooking vegetables, pay attention to heat—not half-cooked, not overcooked or burnt, but just right. This is to preserve various nutrients in vegetables as much as possible. These nutrients are beneficial to eye health, such as vitamins and trace elements.

[53] What should be noted when mixing dough for pasta?

Answer: When mixing dough, alkali and baking soda should be added in the right amount, neither too much nor too little, because many vitamins in pasta are acidified in an acidic environment (such as over-fermentation) and dissolved in an alkaline environment, both of which cause loss of nutritional value. These nutrients are helpful for eye health.

[54] Which cooking method is better for myopia-preventing foods?

Answer: Use steaming and stewing more often. These cooking methods relatively better preserve nutrients in food, allowing food to play a greater role in myopia prevention.

[55] How to pair foods to prevent myopia?

Answer: Prepare nutrient-rich vegetables and diversify meals. Such as brown rice, beans, and green and yellow vegetables like rapeseed, carrot, and bell pepper, as well as radish, cabbage, spinach, kale, and cucumber. In addition, pair with iodine-rich foods such as kelp, fish, and dried small shrimp. These foods not only contain large amounts of eye-beneficial vitamins and trace elements but also contain large amounts of calcium, which plays an important role in the development of the eye sclera and can prevent the occurrence and development of myopia.

References

[1] Bai, W., & Bai, H. (1984). Prevention and Treatment of Myopia (Revised and Expanded Edition). Hunan Science and Technology Press.

[2] Cong, P., & Ma, G. (1989). Myopia and Its Prevention and Treatment. Hubei Science and Technology Press.

[3] Fan, L. (2011). A Comprehensive Collection of Famous Doctors' Medicated Diets. Zhejiang Science and Technology Publishing House.

[4] Gao, J. (2012). Food Therapy Congees and Meals to Cure All Diseases. Shaanxi Science and Technology Press.

[5] Huang, M. (1990). Comprehensive Self-Prevention and Treatment Methods for Adolescent Myopia. South China University of Technology Press.

[6] Jin, M. (2009). Eating This Way Is the Healthiest: Food Therapy Cures All Diseases. Jiuzhou Press.

[7] Li, X., & Wang, L. (2009). Eating This Way Is Best for Health Preservation. Rural Reading Materials Publishing House.

[8] Shi, Y. (2011). Preventing and Treating Myopia Starts at Age Three. Xi'an Jiaotong University Press.

[9] Shi, Y. (2017). Say No to Myopia: Dr. Shi Yining Has Something to Say. Shaanxi Science and Technology Press.

[10] Wang, W. (2011). Nourishing Medicated Diets. Changsha: Hunan Science and Technology Press.

[11] Wang, L., & Wang, J. (2006). What Not to Eat. Jindun Publishing House.

[12] Wu, Y. (2011). Are You Getting the Right Nutrition? A Nutrition PhD Tells You the Nutrition Myths You Must Know. SDX Joint Publishing Company.

[13] Yao, S., Wang, H., & Pan, D. (2011). Complete Book of TCM Health-Preserving Medicated Diets and Food Therapy. Jilin Science and Technology Press.

[14] Zhang, M. (1991). 300 Questions on the Prevention and Treatment of Common Eye Diseases. Jindun Publishing House.

[15] Zhang, T. (2015). Quick Reference on What to Eat and What to Avoid for Common Diseases. Shanghai Popular Science Press.

[16] Zhang, H. (2020). Joyful Food Time: Nutritious Breakfasts Kids Love. China Textile Press Co., Ltd.

[17] Editorial Board of the Illustrated Life · Gourmet World Series (2009). Seeking Food Is Better Than Seeking a Doctor: 1000 Recipes for Preventing and Managing Common Diseases. Shanghai Popular Science Press.

[18] Shi, G., & Shi, Y. (1993). 150 Questions on the Prevention and Treatment of Common Eye Diseases. World Publishing Corporation Xi'an.

[19] Xing, C. (1989). Questions and Answers on the Prevention and Treatment of Common Eye Diseases. Guangdong Science and Technology Press.

[20] Tao, H. (2009). A Complete Guide to Common Eye Diseases. Jilin Science and Technology Press.